

# BHAVAN'S COLLEGE (AUTONOMOUS), ANDHERI-WEST

## PRACTICAL JOURNAL

Class: TYIT

Sem: VI

Roll No.: TYIT25

Date: 25 / 2 / 2025

Course Name: Geographical Information System

Page No.:

Practical Number: 8

### Q.8) Advanced GIS Operations:

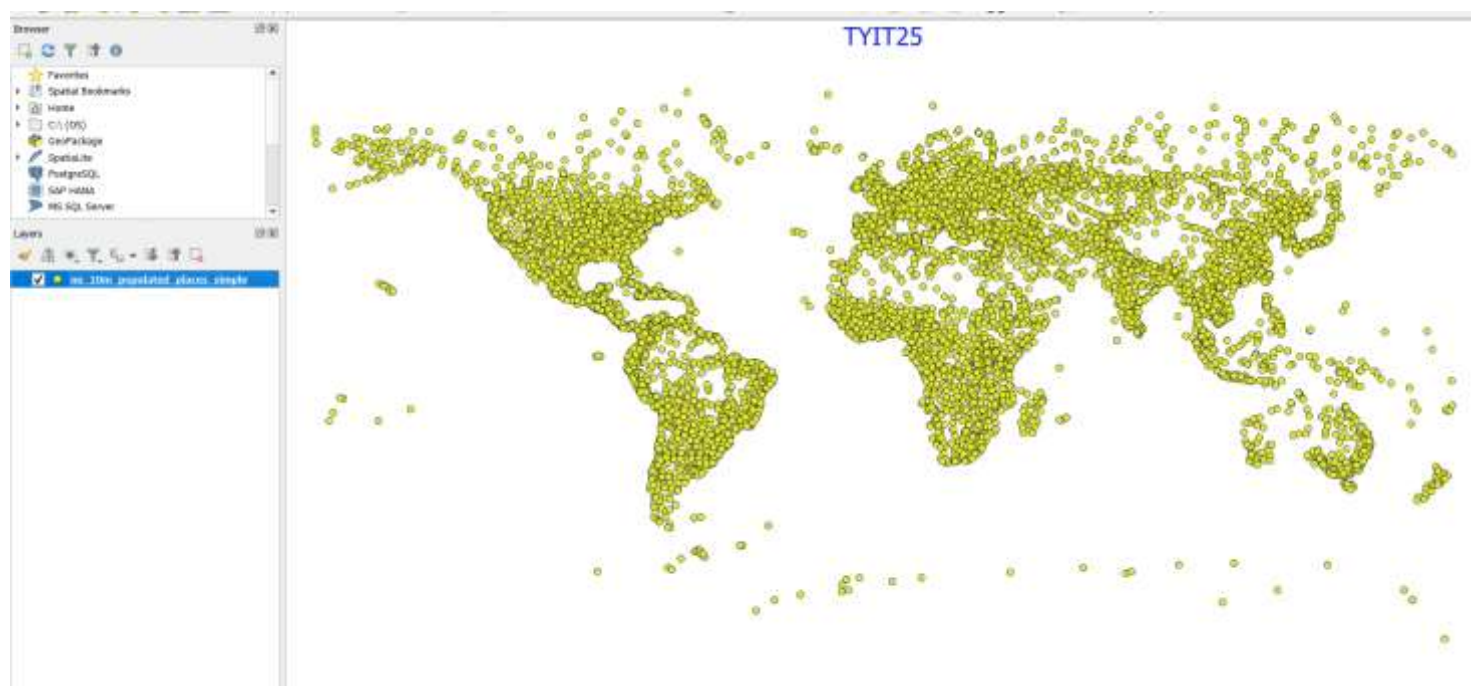
- a) Nearest Neighbor Analysis.
- b) Sampling Raster Data using Points or Polygons.
- c) Interpolating Point Data.

:-

#### a) Nearest Neighbor Analysis.

#### Step 1: Adding a Vector Layer

1. Navigate to **Layer** → **Add Layer** → **Add Vector Layer** → **Add File**.
2. Select the following shapefile:
  - o **Practical\_08\DATA\A\ne\_10m\_populated\_places\_simple\ne\_10m\_populated\_places\_simple.shp**
3. Click **Add**, then **OK**.



#### Step 2: Adding a Delimited Text Layer

1. Navigate to **Layer** → **Add Layer** → **Add Delimited Text Layer**.
2. Select the file:
  - o **Practical\_08\DATA\A\earthquakes\_2021\_11\_25\_14\_31\_59\_+0530.tsv**
3. Ensure all **coordinate fields** are set by default.
4. Click **Add**, then **Close**.

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Data Source Manager | Delimited Text

File name: I217T154402Z-001-20250221T141041Z-001\Practical\_08-20211217T154402Z-001\Practical\_08\DATA\A\earthquakes\_2021\_11\_25\_14\_31\_59\_+0530.tsv

Layer name: earthquakes\_2021\_11\_25\_14\_31\_59\_+0530 Encoding: UTF-8

**File Format**

☐ CSV (comma separated values) ☒ Tab ☐ Colon ☐ Space

☐ Regular expression delimiter ☐ Semicolon ☐ Comma Others:

☒ Custom delimiters Quote: "  Escape: "

**Record and Fields Options**

Number of header lines to discard: 0 ☐ Decimal separator is comma

☒ First record has field names ☐ Trim fields

☒ Detect field types ☐ Discard empty fields

**Custom boolean literals**

True:  False:

**Geometry Definition**

☒ Point coordinates X field: Longitude Z field:

☐ Well known text (WKT) Y field: Latitude M field:

☐ No geometry (attribute only table) ☐ DMS coordinates

Geometry CRS: EPSG:4326 - WGS 84

**Layer Settings**

**Sample Data**

| Search Parameters | Year | Mn | Dv | Hr | Mn |
|-------------------|------|----|----|----|----|
|-------------------|------|----|----|----|----|

Close Add Help

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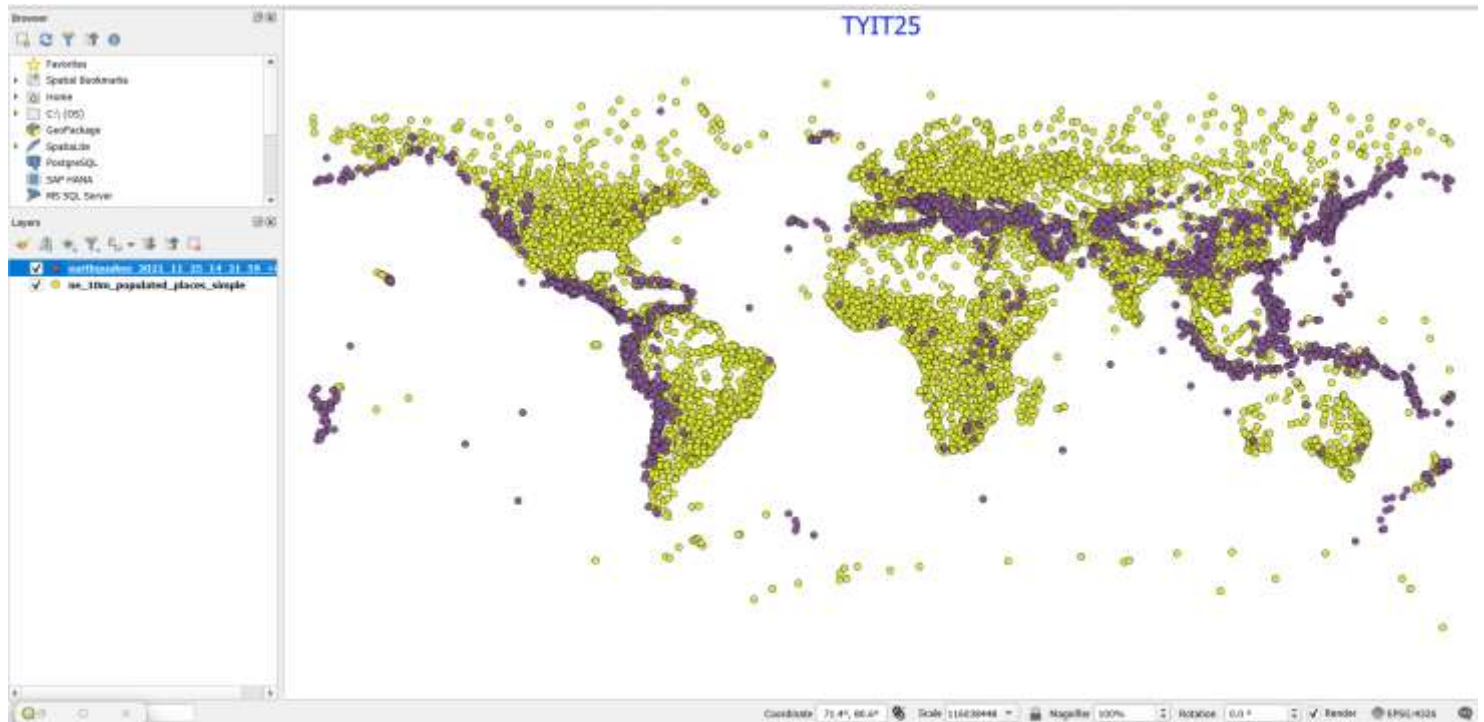
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In the **Earthquake Layer**, upon opening the **Attribute Table**, we observe a total of **2,586 features**. However, some records contain **NULL values** in the **Longitude** and **Latitude** fields. To ensure data accuracy, these entries must be removed.

| earthquakes_2021_11_25_14_31_59_+0530 — Features Total: 2586, Filtered: 2586, Sel... |      |                           |          |           |                  |  |
|--------------------------------------------------------------------------------------|------|---------------------------|----------|-----------|------------------|--|
|                                                                                      | Vol  | Location Name             | Latitude | Longitude | Focal Depth (km) |  |
| 1                                                                                    | NULL | NULL                      | NULL     | NULL      | NULL             |  |
| 2                                                                                    | 5459 | INDONESIA: GA...          | -0.03    | 127.25    | NULL             |  |
| 3                                                                                    | NULL | JAPAN: SEA OF JAPAN       | 36.5     | 133.5     | NULL             |  |
| 4                                                                                    | NULL | INDONESIA: JAVA: SUKABUMI | -6.84    | 106.96    | NULL             |  |
| 5                                                                                    | NULL | MEXICO                    | 20       | -105      | 10               |  |
| 5                                                                                    | NULL | MEXICO: NEAR ...          | 20       | -105      | 60               |  |
| 7                                                                                    | NULL | VENEZUELA: CA...          | 10.3     | -63.3     | NULL             |  |
| 8                                                                                    | NULL | COSTA RICA                | 10       | -85.5     | 60               |  |

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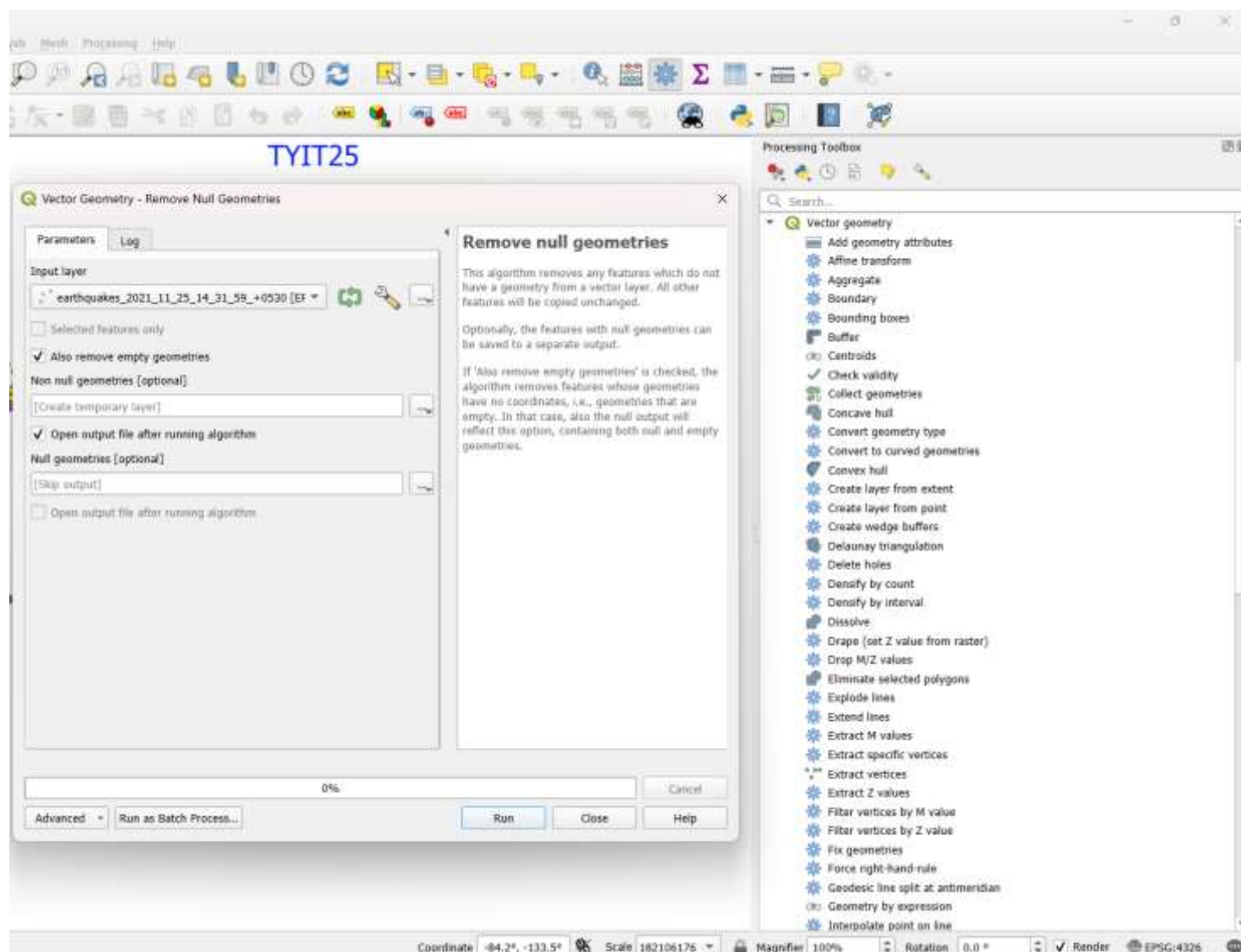
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### Step 3: Removing Null and Empty Geometries

1. Open the **Processing Toolbox**.
2. Navigate to **Vector Geometry** → **Remove Null Geometries**.
3. Check the option "**Also remove empty geometries**".
4. Click **Run**, then wait for the process to complete.
5. Close the tool once finished.

This ensures that all features with missing or empty geometries are removed from the dataset.



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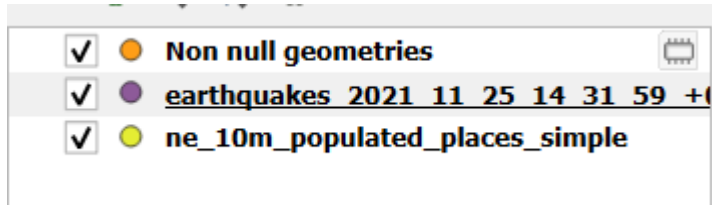
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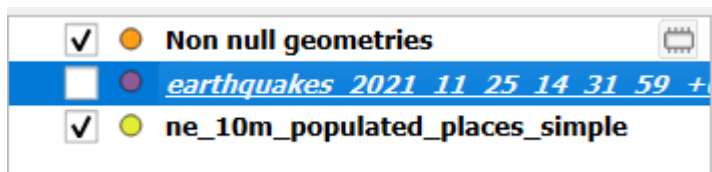
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Deselecting the Earthquake Layer



Upon opening the **Attribute Table** of the **Non-Null Geometries** layer, we can confirm that the **Latitude** and **Longitude** fields no longer contain any **NULL** values.

Non null geometries — Features Total: 2578, Filtered: 2578, Selected: 0

|   | Vol  | Location Name             | Latitude | Longitude | Focal Depth (km) |
|---|------|---------------------------|----------|-----------|------------------|
| 1 | 5459 | INDONESIA: GA...          | -0.03    | 127.25    | NULL             |
| 2 | NULL | JAPAN: SEA OF JAPAN       | 36.5     | 133.5     | NULL             |
| 3 | NULL | INDONESIA: JAVA: SUKABUMI | -6.84    | 106.96    | NULL             |
| 4 | NULL | MEXICO                    | 20       | -105      | 10               |
| 5 | NULL | MEXICO: NEAR ...          | 20       | -105      | 60               |
| 6 | NULL | VENEZUELA: CA...          | 10.3     | -63.3     | NULL             |
| 7 | NULL | COSTA RICA                | 10       | -85.5     | 60               |
| 8 | NULL | TURKEY: KARS,...          | 40.3     | 43.1      | 60               |

Show All Features

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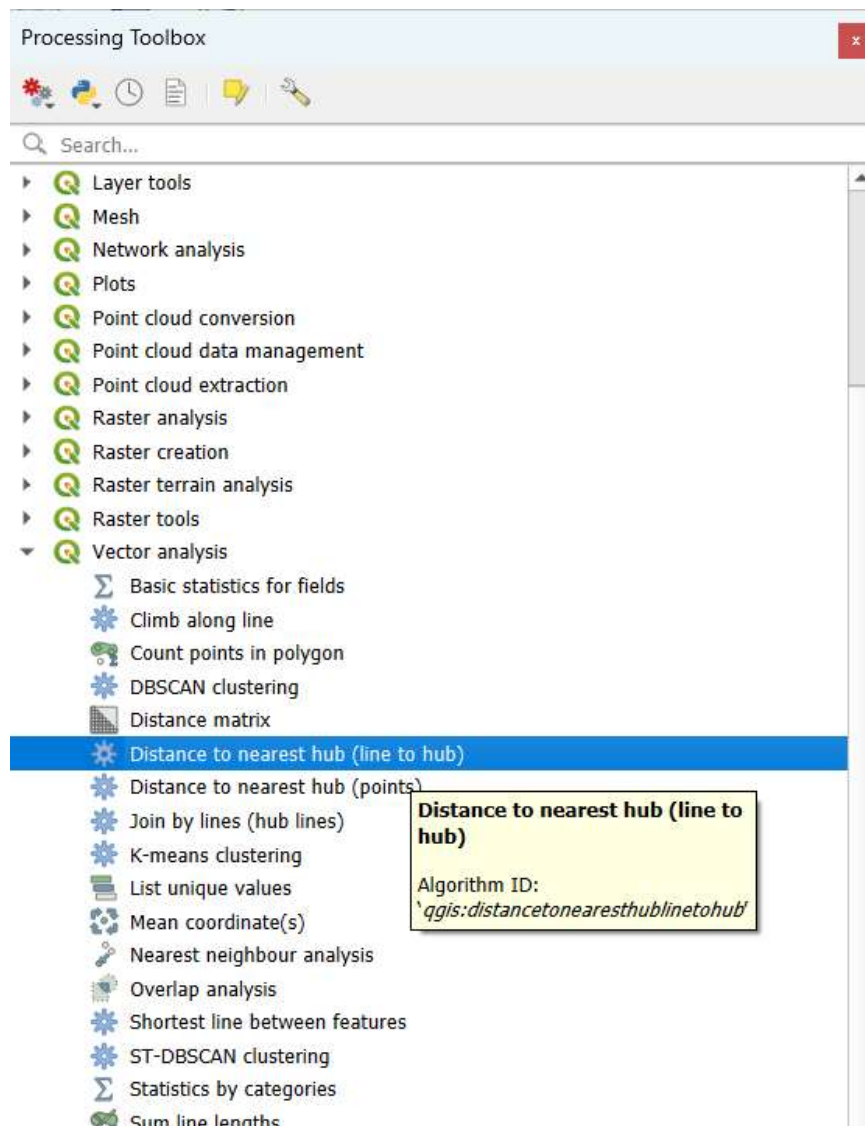
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### Step 4: Calculating Distance to the Nearest Hub

1. Navigate to **Processing** → **Toolbox**.
2. Expand **Vector Analysis**.
3. Select **Distance to Nearest Hub (Line to Hub)**.

This tool calculates the shortest distance between each point in one layer to its nearest feature in another layer, typically used for proximity analysis.



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### Step 5: Configuring Parameters for Distance to Nearest Hub Analysis

1. In the **Distance to Nearest Hub (Line to Hub)** tool, set the following parameters:
  - **Source Point Layer:** Select **Non-Null Geometries**.
  - **Destination Hubs Layer:** Choose **ne\_10m\_populated\_places\_simple**.
  - **Hub Layer Name Attribute:** Set to **Name**.
  - **Measurement Unit:** Select **Kilometers**.
  - **Hub Distance:** Save the output file to a preferred location.
2. Enable **"Open Attribute Table After Running Algorithm"**.
3. Click **Run**, then wait for the process to complete.
4. Review the output data in the attribute table.

The screenshot shows the 'Distance to Nearest Hub (Line to Hub)' dialog box in QGIS. The 'Parameters' tab is active, showing the following settings:

- Source points layer:** Non null geometries [EPSG:4326] (Selected features only is unchecked).
- Destination hubs layer:** ne\_10m\_populated\_places\_simple [EPSG:4326] (Selected features only is unchecked).
- Hub layer name attribute:** abc name.
- Measurement unit:** Kilometers.
- Hub distance:** C:/Users/jaymh/OneDrive/Desktop/Earth\_Quake8.gpkg (Open output file after running algorithm is checked).

The right pane contains the following text:

**Distance to nearest hub (line to hub)**

Given an origin and a destination layers, this algorithm computes the distance between origin features and their closest destination one. Distance calculations are based on the features center.

The resulting layer contains lines linking each origin point with its nearest destination feature.

The bottom of the dialog shows a progress bar at 0%, a 'Cancel' button, and a 'Run' button. There are also 'Advanced', 'Run as Batch Process...', 'Close', and 'Help' buttons.

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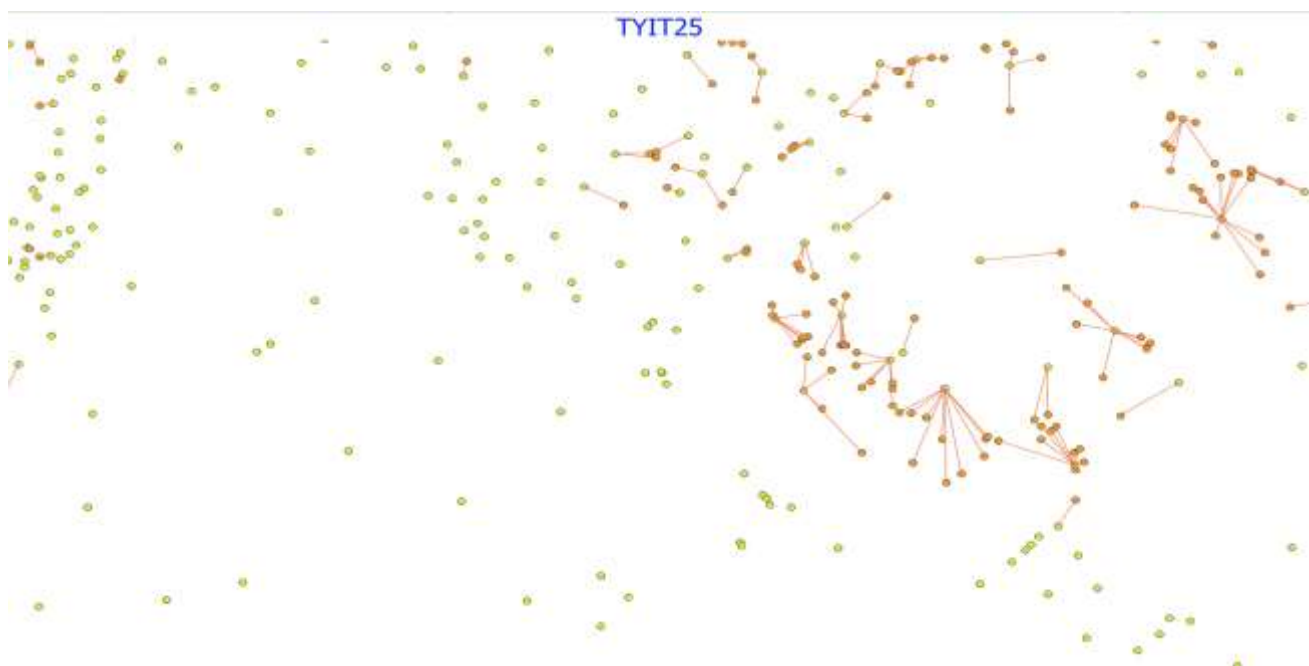
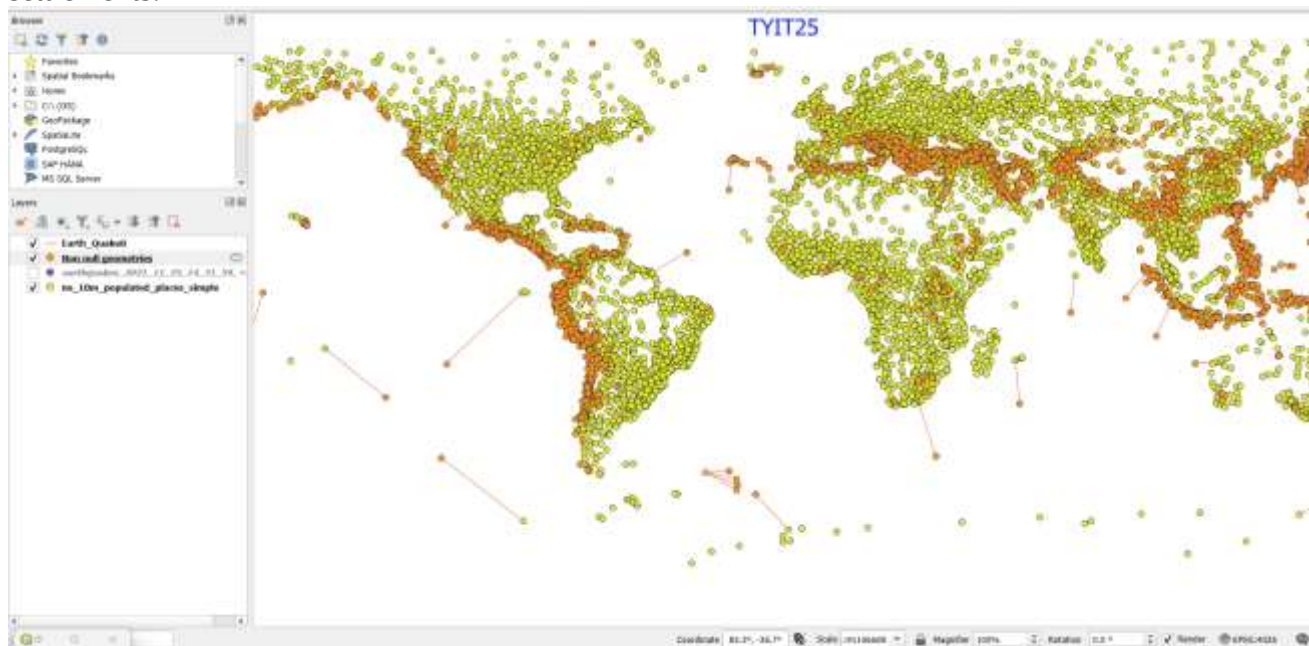
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### Final Output of Practical 08 - Section A:

- **Orange Dots** represent **Earthquake locations**.
- **Yellow Dots** indicate **Populated Places**.
- **Connecting Lines** illustrate the **nearest hub distances** between earthquakes and the closest populated places.

This visualization effectively demonstrates the spatial relationship between earthquake occurrences and human settlements.



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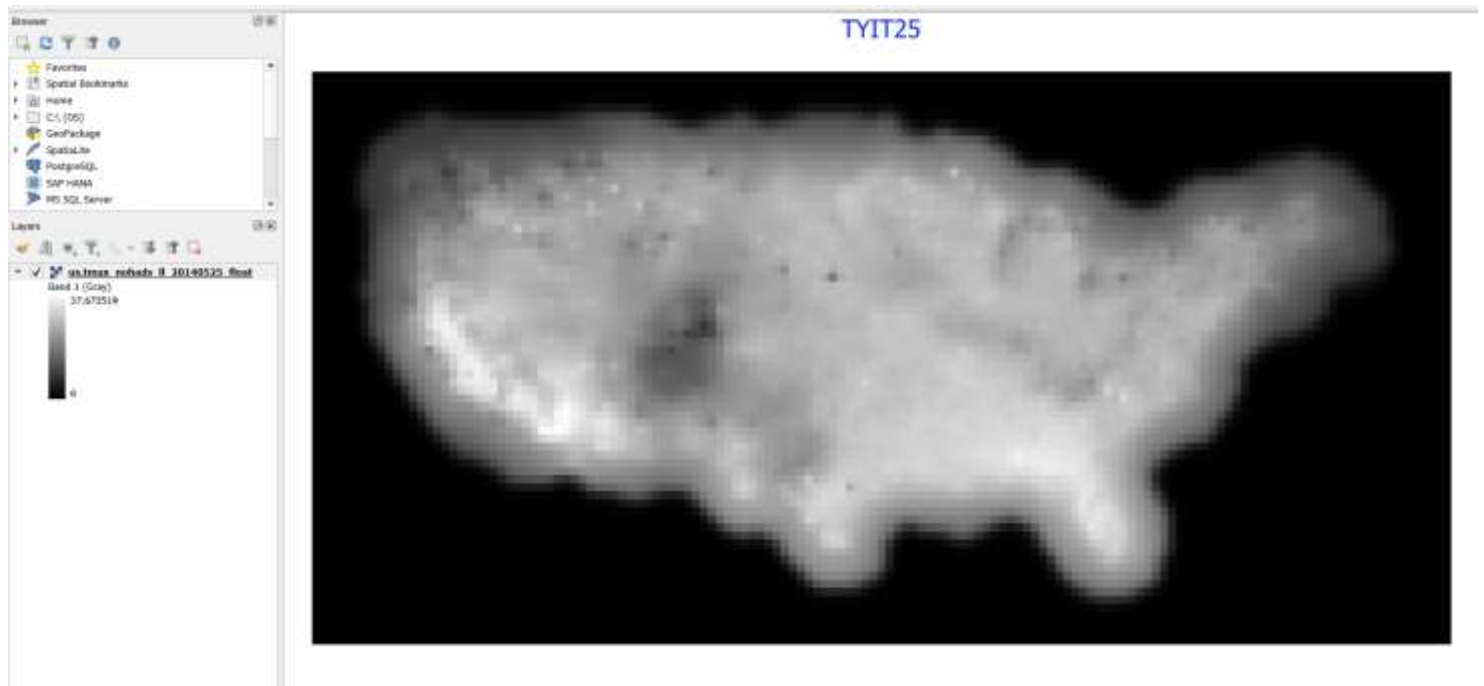
### b) Sampling Raster Data using Points or Polygons.

:-

#### Step 6: Adding a Raster Layer

1. Navigate to **Layer** → **Add Layer** → **Add Raster Layer**.
2. Select the file:
  - o **Practical\_08\DATA\B\us.tmax\_nohads\_ll\_20140525\_float.tif**
3. Click **Add**, then **OK**.

This step loads a **.TIF raster file**, which likely represents temperature or climate data.



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### Step 7: Adding a Delimited Text Layer

1. Navigate to **Layer** → **Add Layer** → **Add Delimited Text Layer** → **Add File**.
2. Select the file:
  - **Practical\_08\DATA\B\2013\_Gaz\_ua\_national\2013\_Gaz\_ua\_national.txt**
3. In **File Format**, select **Tab**.
4. Under **Record and Fields Options**, keep the default settings.
5. In **Geometry Definition**, set **Point Coordinate** as:
  - **X Field:** INPUTLONG (Longitude)
  - **Y Field:** INPUTLAT (Latitude)
6. Ensure all coordinates are set **automatically**.
7. Click **Add**, then **Close**.

This step integrates a text-based dataset containing **geographic coordinates** into the project.

The screenshot shows the 'Data Source Manager | Delimited Text' dialog box. The 'File name' field contains the path: '1771544022-001-20250221T141041Z-001\Practical\_08-20211217T154402Z-001\Practical\_08\DATA\B\2013\_Gaz\_ua\_national\2013\_Gaz\_ua\_national.txt'. The 'Layer name' is '2013\_Gaz\_ua\_national' and the 'Encoding' is 'UTF-8'. The 'File Format' section has 'Tab' selected. The 'Record and Fields Options' section has 'First record has field names' and 'Detect field types' checked. The 'Geometry Definition' section has 'Point coordinates' selected, with 'X field' set to 'INTPTLONG' and 'Y field' set to 'INTPTLAT'. The 'Layer Settings' section is partially visible at the bottom.

| Field                     | Value                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| File name                 | 1771544022-001-20250221T141041Z-001\Practical_08-20211217T154402Z-001\Practical_08\DATA\B\2013_Gaz_ua_national\2013_Gaz_ua_national.txt |
| Layer name                | 2013_Gaz_ua_national                                                                                                                    |
| Encoding                  | UTF-8                                                                                                                                   |
| File Format               | Tab                                                                                                                                     |
| Record and Fields Options | First record has field names, Detect field types                                                                                        |
| Geometry Definition       | Point coordinates, X field: INTPTLONG, Y field: INTPTLAT                                                                                |

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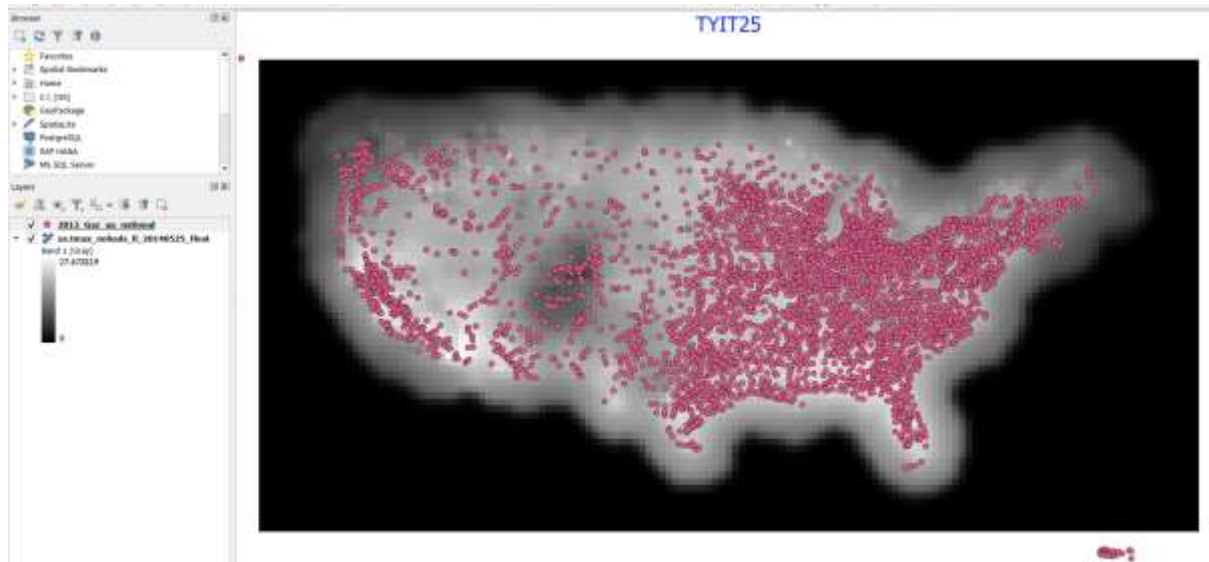
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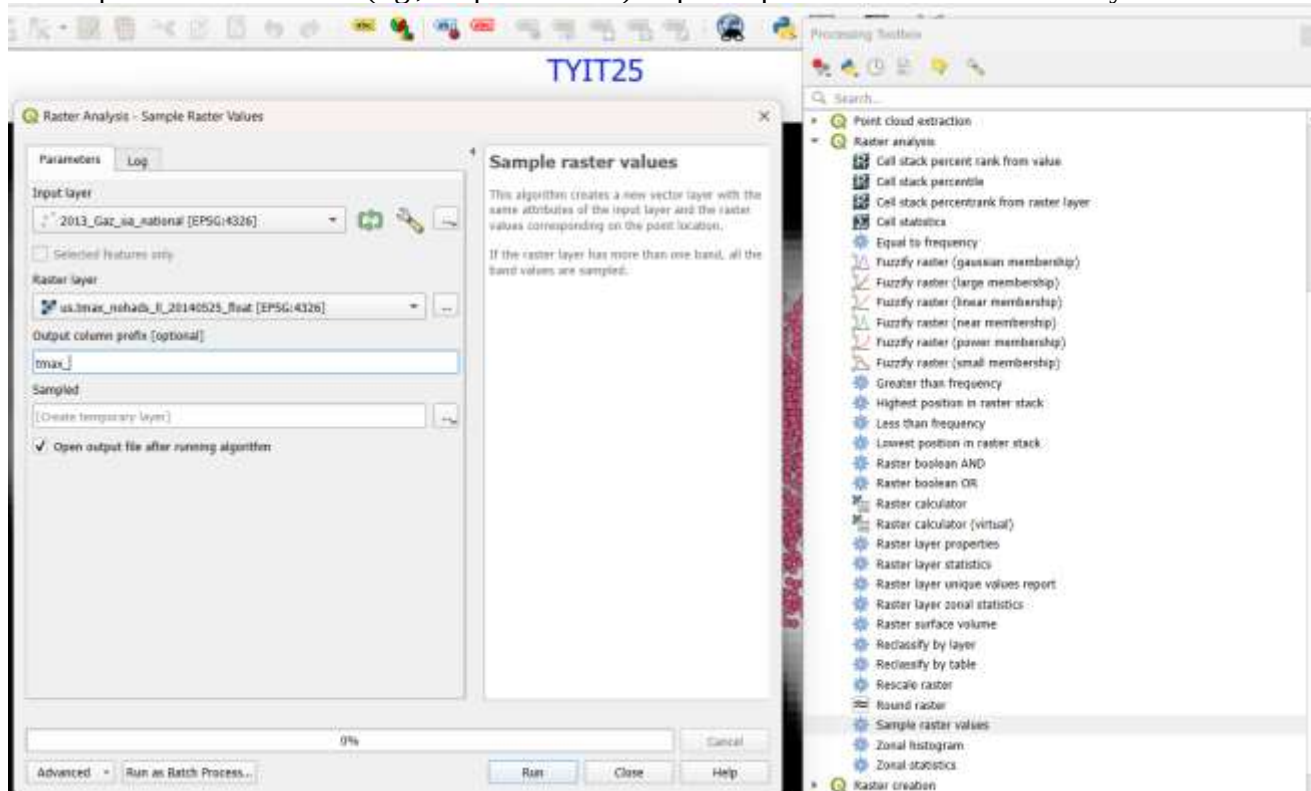
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### Step 8: Sampling Raster Values

1. Navigate to **Processing** → **Toolbox** → **Raster Analysis** → **Sample Raster Values**.
2. In **Parameters**, configure the following:
  - **Raster Layer:** Select **us.tmax\_nohads**.
  - **Output Column Prefix:** Set as **tmax\_**.
3. Click **Run**, then **Close** once the process is complete.

This step extracts raster values (e.g., temperature data) at specific point locations for further analysis.



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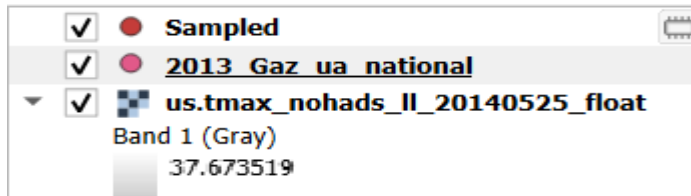
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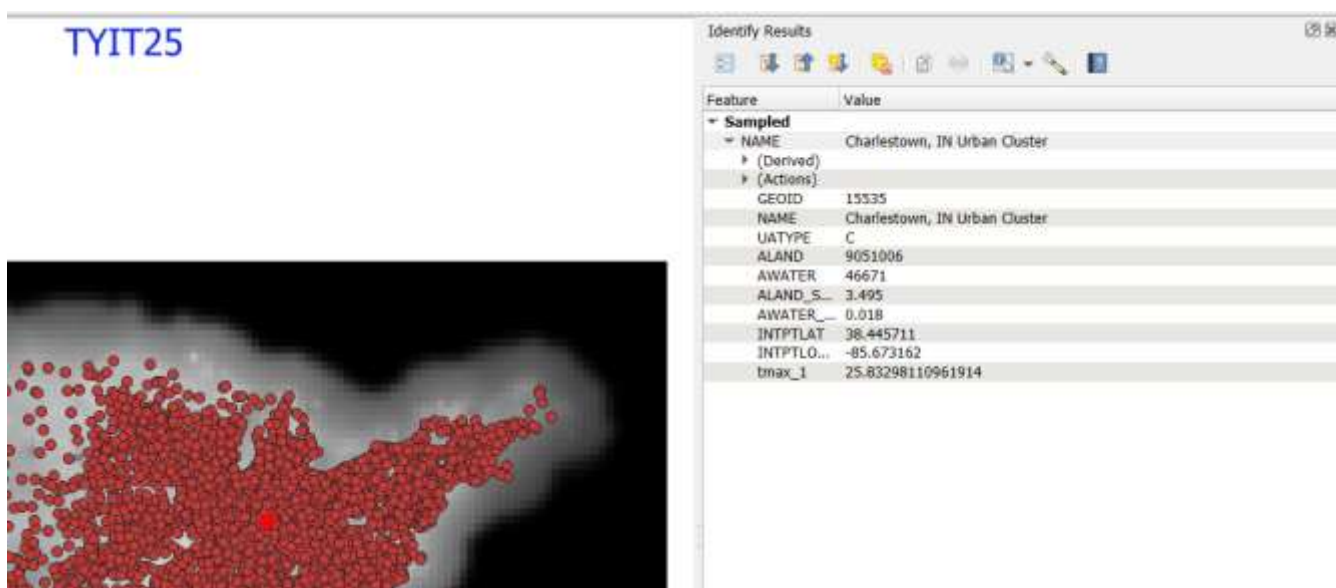
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When using the **Identify Feature** tool to click on any area of the map, the **tmax\_1** attribute becomes visible, displaying the extracted raster value for that location.



Additionally, when we **right-click** on the **sampled layer** and select **Open Attribute Table**, we can see that the **tmax\_1** column has been successfully added, containing the extracted raster values.

Sampled — Features Total: 3601, Filtered: 3601, Selected: 0

|   | AWATER  | ALAND_SQMI | AWATER_SQMI | INTPTLAT  | INTPTLONG   | tmax_1             |
|---|---------|------------|-------------|-----------|-------------|--------------------|
| 1 | 300497  | 11.283     | 0.116       | 29.967224 | -92.096238  | 30.44220924377...  |
| 2 | 19786   | 4.355      | 0.008       | 34.179237 | -82.379726  | 29.73669052124...  |
| 3 | 13221   | 2.071      | 0.005       | 44.948612 | -90.315875  | 26.87336730957...  |
| 4 | 52820   | 2.864      | 0.02        | 33.824742 | -88.554591  | 31.27118110656...  |
| 5 | 120864  | 12.789     | 0.047       | 45.463186 | -98.471033  | 28.48344421386...  |
| 6 | 1929686 | 15.443     | 0.745       | 46.976365 | -123.796056 | 28.483444213867188 |
| 7 | 9826393 | 131.137    | 3.794       | 39.508976 | -76.303431  | 26.72383880615...  |
| 8 | 12572   | 1.181      | 0.005       | 33.830523 | -101.846278 | 21.67542076110...  |

Show All Features

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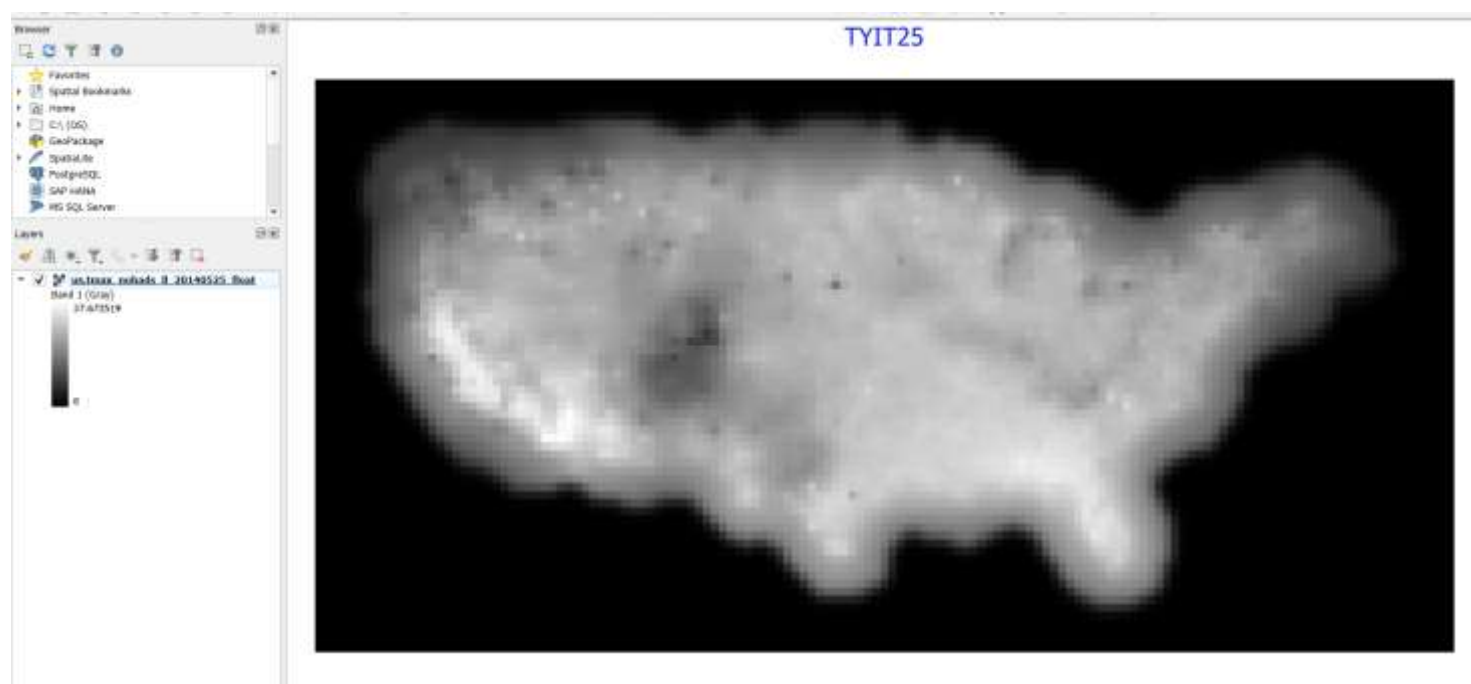
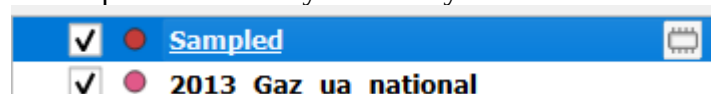
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### Step 9: Removing Unnecessary Layers

1. In the **Layers Panel**, locate the following layers:
  - **Sampled**
  - **2013\_Gaz\_ua\_national**
2. **Right-click** on each layer and select **Remove Layer**.
3. Click **Yes** to confirm the removal.

This step ensures that only relevant layers remain for further analysis.



### Step 10: Adding a Shapefile Layer

1. Navigate to **Layer → Add Layer → Add Vector Layer**.
2. Select the **.shp** file:
  - **Practical\_08\DATA\B\tl\_2018\_us\_county\tl\_2018\_us\_county.shp**
3. Click **Add**, then **Close**.

This step adds a vector layer representing **U.S. county boundaries** for further spatial analysis.

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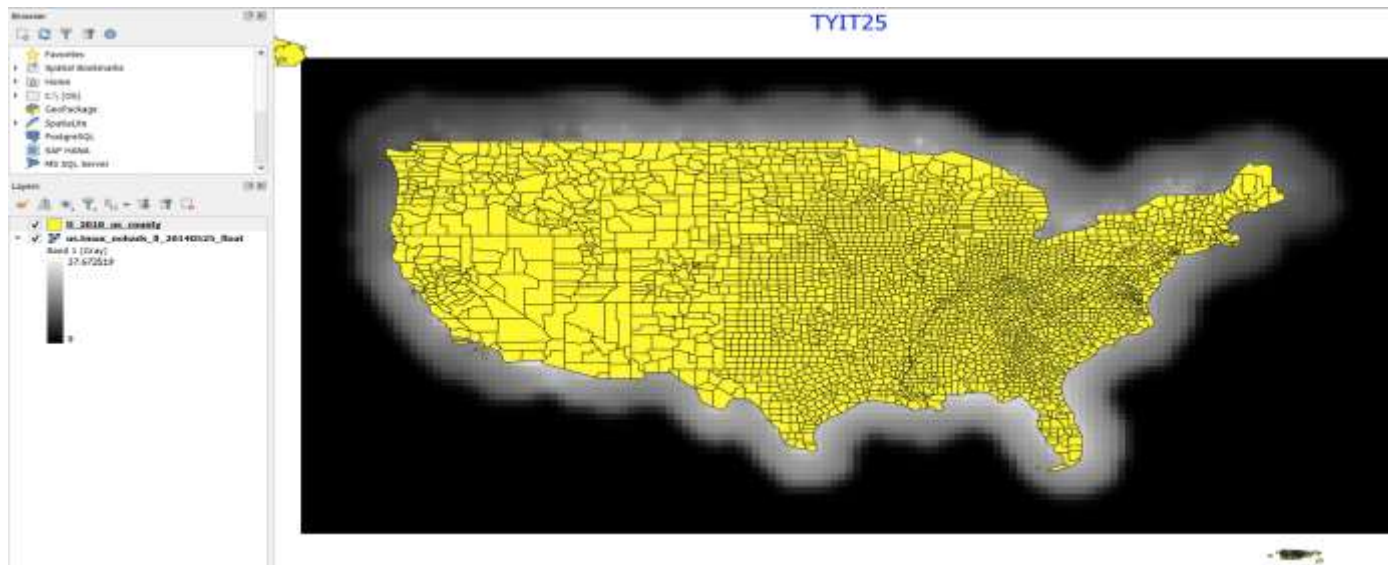
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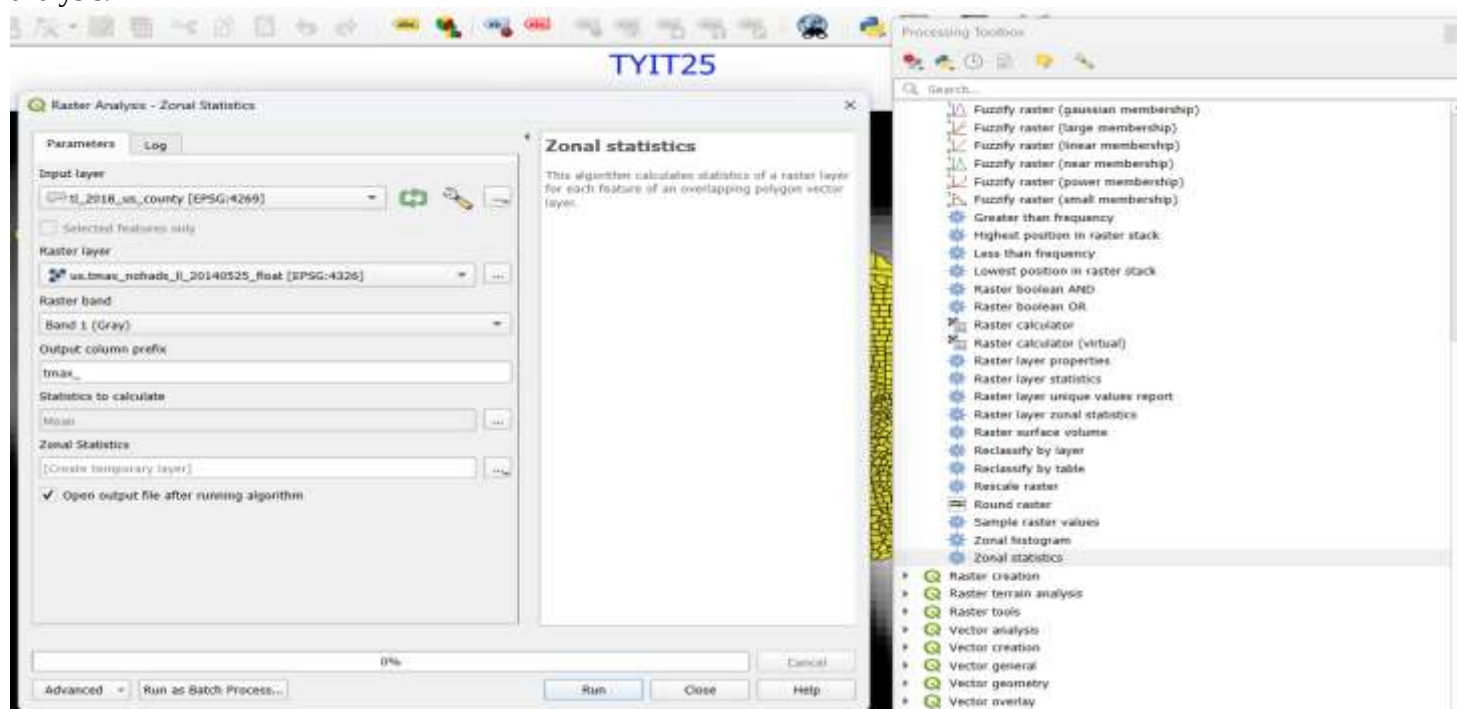
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### Step 11: Performing Zonal Statistics Analysis

1. Navigate to **Processing** → **Toolbox** → **Raster Analysis** → **Zonal Statistics**.
2. In **Parameters**, set the following:
  - **Raster Layer:** Select **us.tmax\_nohads**.
  - **Output Prefix:** Set as **tmax\_**.
  - **Statistics to Calculate:** Select **Mean** only.
3. Click **Run**, then **Close** once the process is complete.

This step computes the **mean temperature values** for each defined zone in the **county shapefile**, enabling regional climate analysis.



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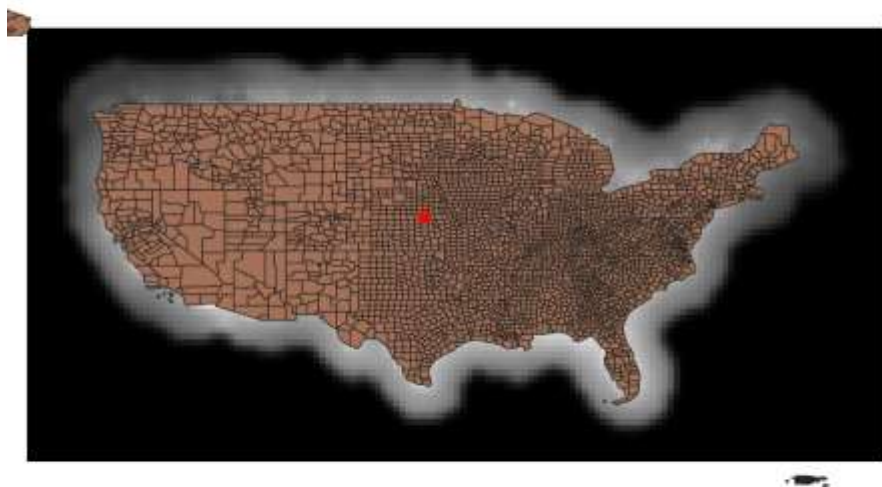
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When using the **Identify Feature** tool to click on any area of the map, the **tmax\_mean** attribute is displayed, showing the maximum temperature value extracted for that specific location.

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| Feature                 | Value            |
|-------------------------|------------------|
| <b>Zonal Statistics</b> |                  |
| NAME                    | Republic         |
| (Derived)               |                  |
| (Active)                |                  |
| STATEFP                 | 20               |
| COUNTYFP                | 137              |
| COUNTYNS                | 00465042         |
| GEOID                   | 20157            |
| NAME                    | Republic         |
| NAMLSAD                 | Republic County  |
| LSAD                    | 08               |
| CLASSFP                 | H1               |
| MTHCC                   | G4020            |
| CSAFP                   | NULL             |
| CRSAFP                  | NULL             |
| METDIVFP                | NULL             |
| FUNCTST                 | A                |
| ALAND                   | 185798868        |
| AWATER                  | 7951912          |
| INTPTLAT                | +36.8288661      |
| INTPTLON                | -95.6508639      |
| tmax_mean               | 27.2351231250405 |

Additionally, when we **right-click** on the **Zonal Statistics** layer and select **Open Attribute Table**, we can see that the **tmax\_mean** column has been successfully added, displaying the calculated mean temperature values for each zone.

| Zonal Statistics — Features Total: 3233, Filtered: 3233, Selected: 0 |            |          |             |              |                   |  |
|----------------------------------------------------------------------|------------|----------|-------------|--------------|-------------------|--|
| JNCSTAT                                                              | ALAND      | AWATER   | INTPTLAT    | INTPTLON     | tmax_mean         |  |
| 1                                                                    | 1477652222 | 10690952 | +41.9158651 | -096.7885168 | 27.64390463907... |  |
| 2                                                                    | 680956809  | 61588406 | +46.2946377 | -123.4244583 | 16.29318978219... |  |
| 3                                                                    | 6016819484 | 29089486 | +34.3592729 | -104.3686961 | 22.33529345194... |  |
| 4                                                                    | 2169287528 | 22832516 | +40.7835474 | -096.6886584 | 28.22672023686... |  |
| 5                                                                    | 1489645187 | 1718484  | +40.1764918 | -098.0468422 | 27.11965577339... |  |
| 6                                                                    | 87748364   | 32509    | +18.1871483 | -065.8711890 | NULL              |  |
| 7                                                                    | 2089691730 | 18198496 | +43.6674723 | -096.7957261 | 27.16365457094... |  |
| 8                                                                    | 2336237985 | 613559   | +30.8852677 | -099.8588613 | 21.17889367956... |  |

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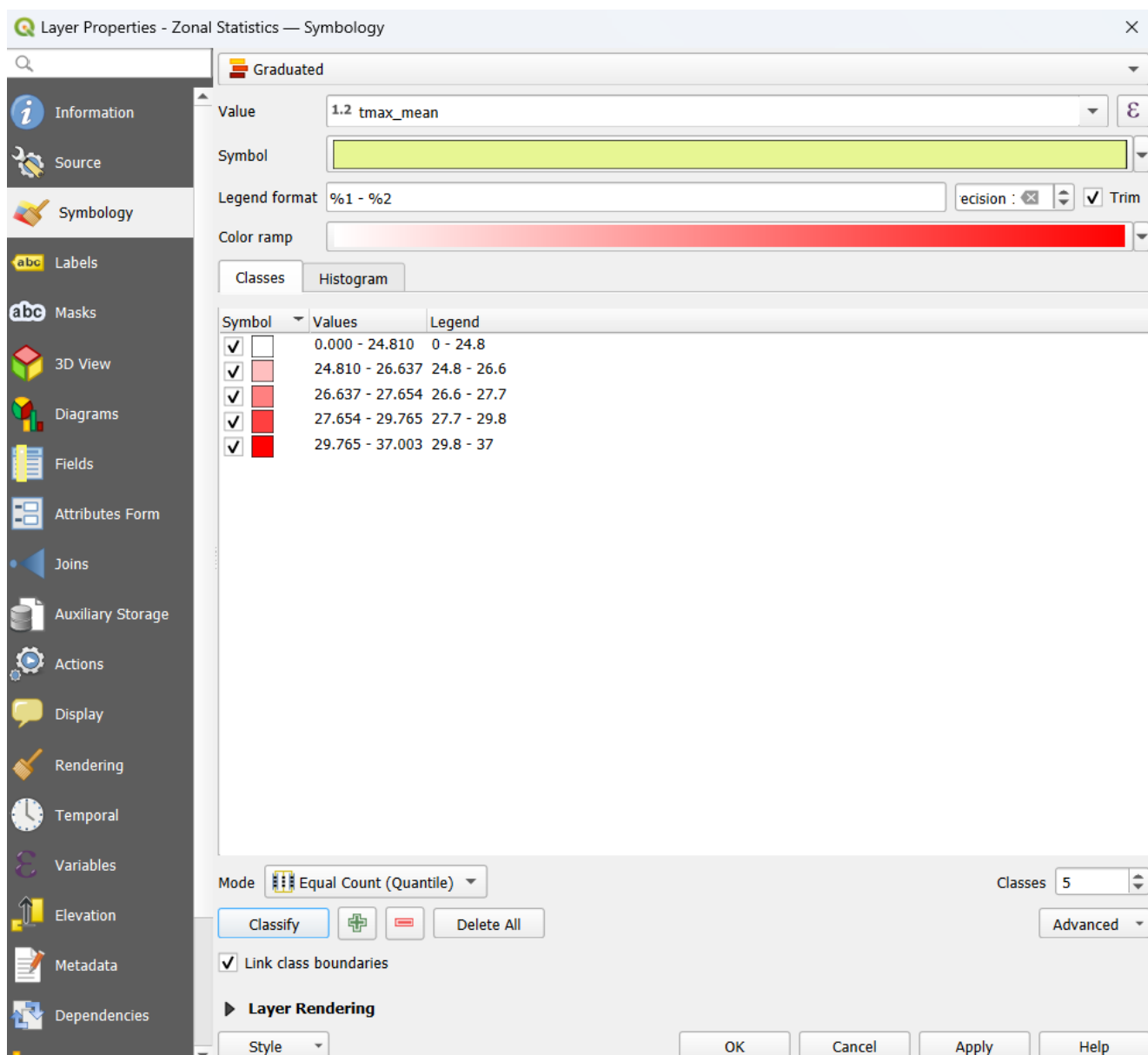
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### Step 12: Applying Symbolology to Zonal Statistics Layer

1. **Right-click** on the **Zonal Statistics** layer and select **Properties**.
2. Navigate to the **Symbolology** tab.
3. Under **Representation Type**, select **Graduated**.
4. Set **Value** to **tmax\_mean**.
5. Customize the **Symbol** and **Color Ramp** based on your preferences.
6. Click on **Classify** to generate value ranges.
7. Click **Apply**, then **OK** to finalize the changes.

This step enhances the visualization by representing the mean temperature values using a **graduated color scale**.



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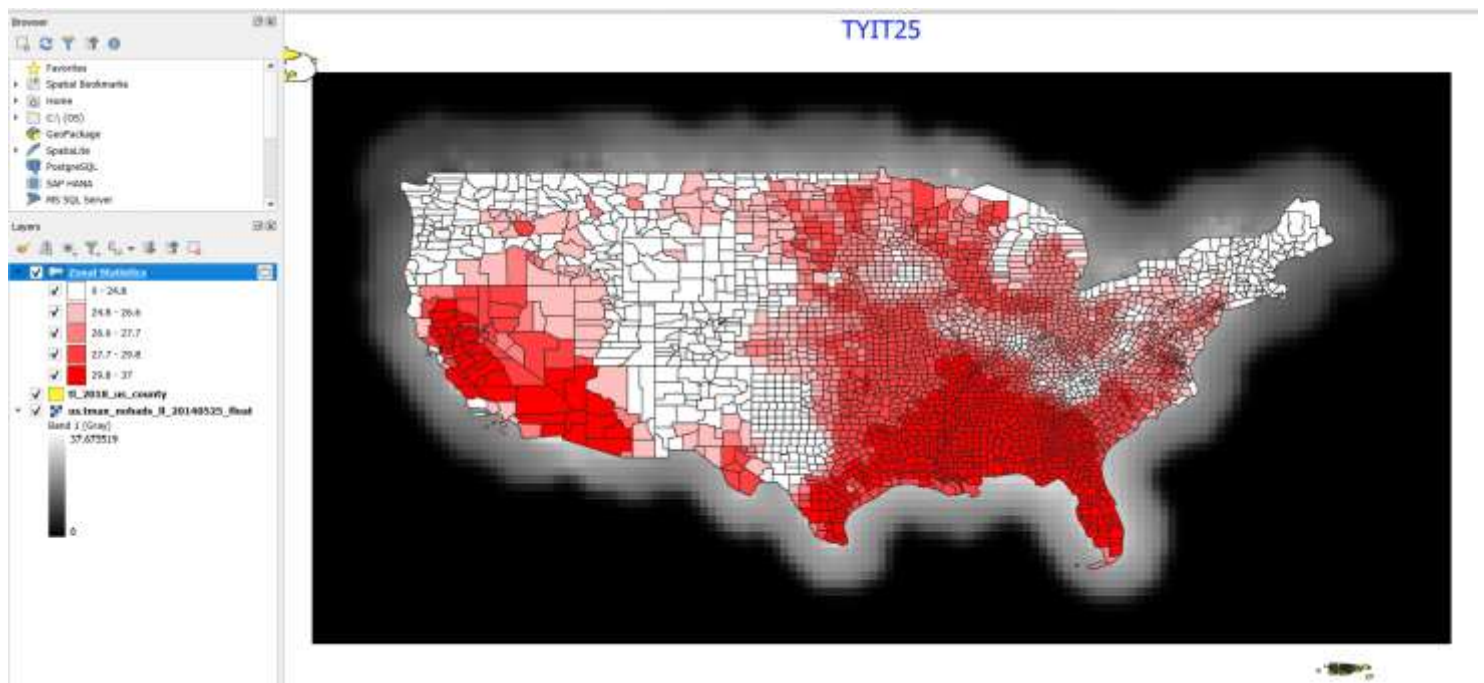
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### Final Output of Practical 08 - Section B:

- The map now visualizes temperature data with a **graduated color scale**, representing the **mean temperature (tmax\_mean)** for different regions.
- **Darker or lighter shades** indicate varying temperature intensities, depending on the chosen symbology.
- Clicking on any area with the **Identify Feature** tool displays the **tmax\_max** and **tmax\_mean** attributes, providing detailed temperature statistics.

This final output effectively demonstrates the spatial distribution of temperature variations across different counties.



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### c) Interpolating Point Data.

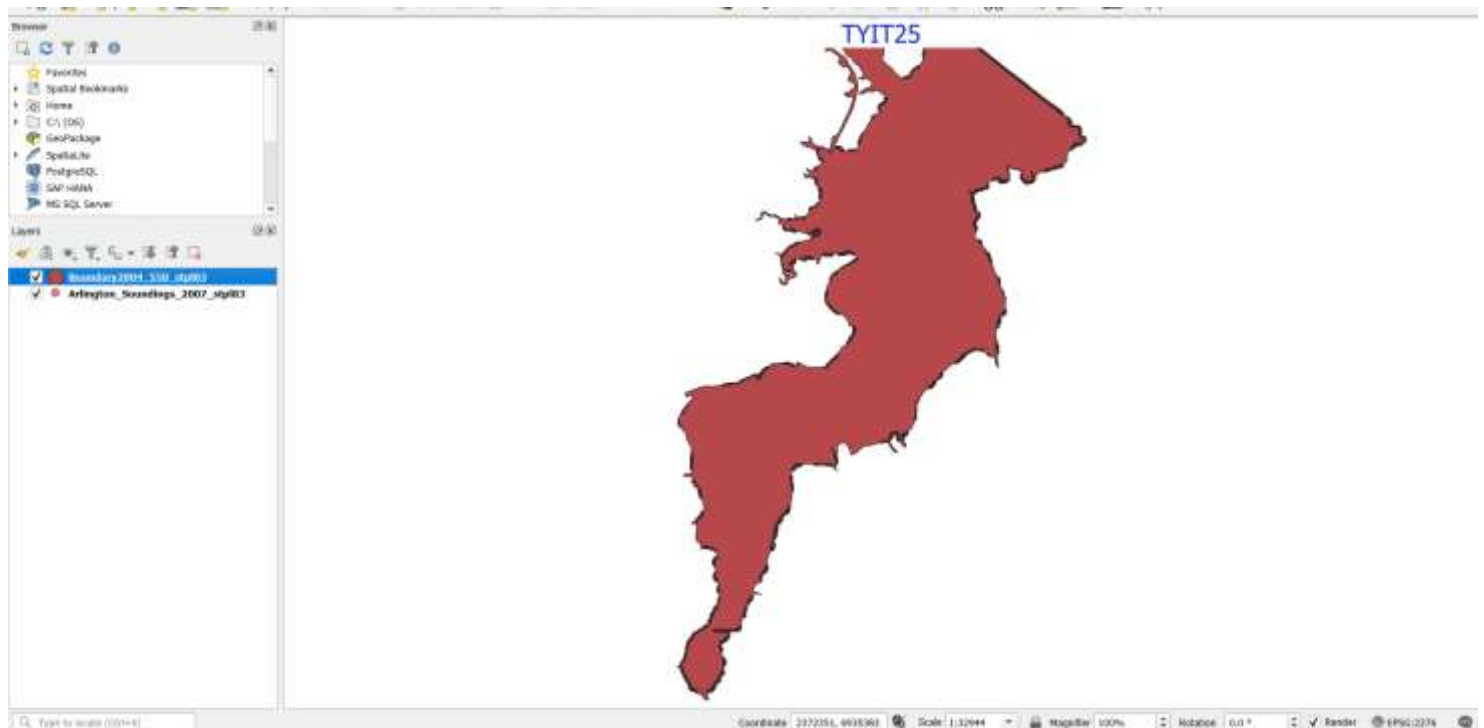
:-

#### Step 13: Adding Shapefile Layers

1. Navigate to **Layer** → **Add Layer** → **Add Vector Layer**.
2. Select the following **.shp** files:
  - **Practical\_08\DATA\C\Shapefiles\Arlington\_Soundings\_2007\_stpl83.shp**
  - **Practical\_08\DATA\C\Shapefiles\Boundary2004\_550\_stpl83.shp**
3. Click **Add**, then **Close**.

This step adds two vector layers:

- **Arlington Soundings 2007**, likely representing depth measurements or soundings.
- **Boundary 2004**, which may define a geographic boundary for analysis.



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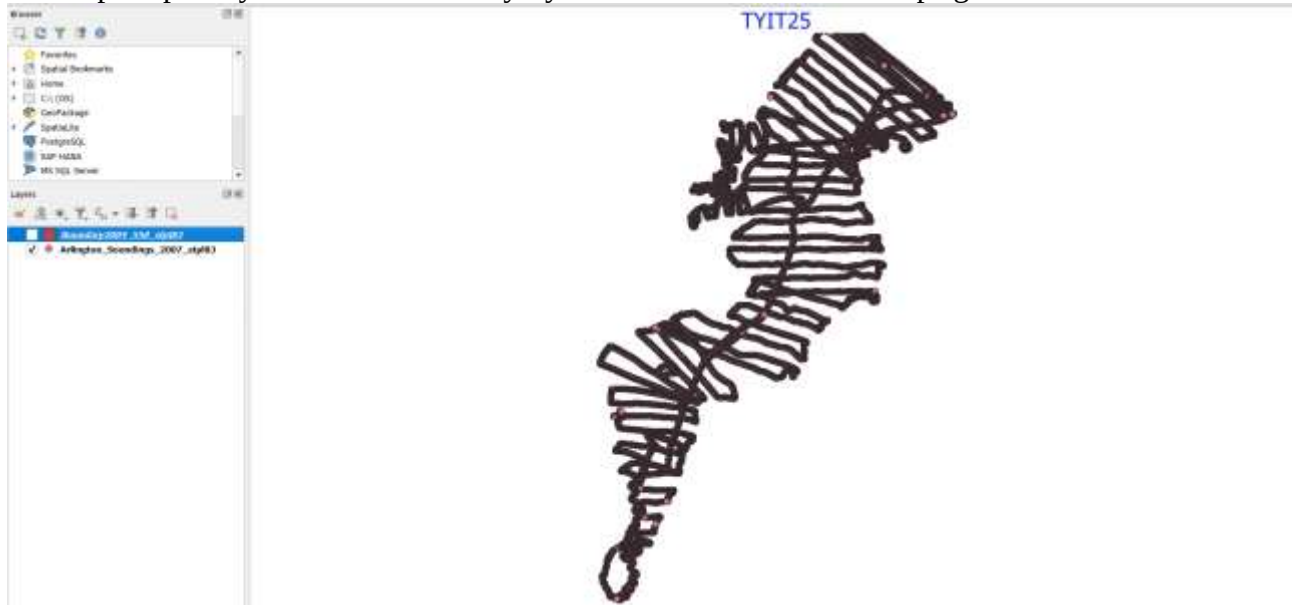
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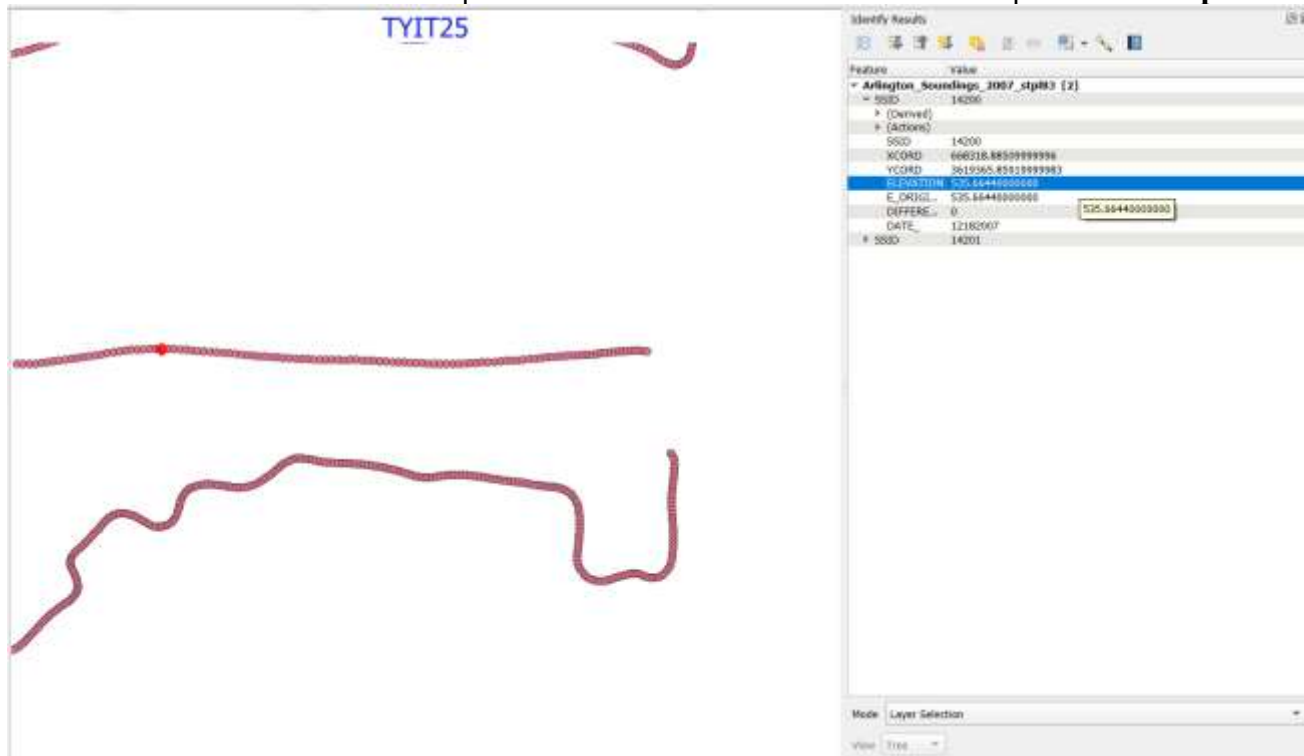
### Step 14: Deselecting the Boundary Layer

1. In the **Layers Panel**, locate the layer **Boundary2004\_550**.
2. **Uncheck** the checkbox next to it to **deselect** or hide the layer from the map view.

This step temporarily removes the boundary layer from visualization while keeping it available for future use.



We need to **zoom in** on the map to visualize the individual points. Using the **Identify Features** tool, we can examine the **Elevation** values associated with these points. These elevation values will be used as input for the **Interpolation** process.



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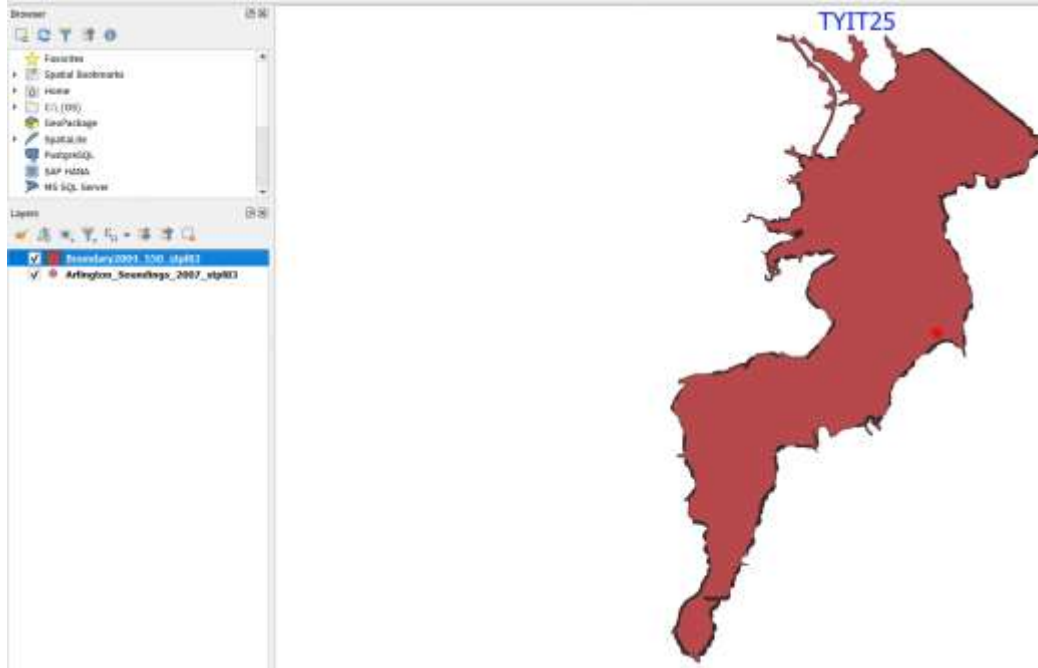
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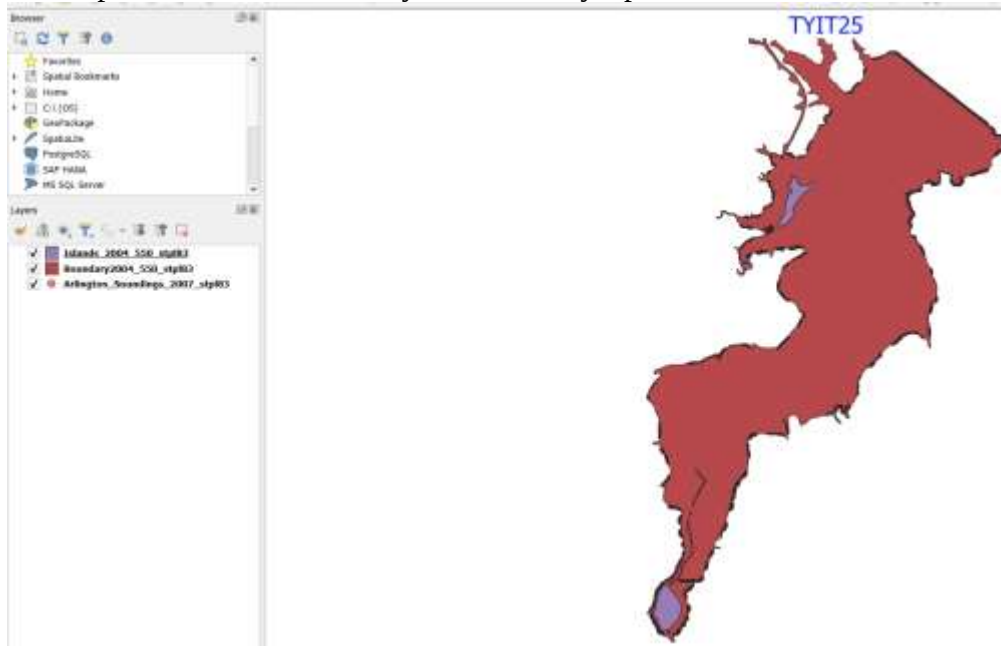
We need to **zoom out** on the map and **reselect** the **Boundary2004\_550** layer to make it visible again.



### Step 15: Adding the Islands Shapefile Layer

1. Navigate to **Layer** → **Add Layer** → **Add Vector Layer**.
2. Select the **.shp** file:
  - o **Practical\_08\DATA\C\Shapefiles\Islands\_2004\_550\_stpl83.shp**
3. Click **Add**, then **Close**.

This step adds the **Islands 2004** layer, which likely represents island boundaries within the study area.



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### Step 16: Performing TIN Interpolation

1. Navigate to **Processing** → **Toolbox** → **Interpolation** → **TIN Interpolation**.
2. **Set the input parameters:**
  - **Vector Layer:** Select **Arlington\_Soundings\_2007\_stpl83**.
  - **Interpolation Attribute:** Select **ELEVATION**.
  - Click on the **Add**  **button** to include this layer.

Then Again,

- **Vector Layer:** Select **Islands\_2004\_550\_stpl83**.
  - **Interpolation Attribute:** Select **ELEVATION**.
  - Click on the **Add**  **button** again.
3. **Choose the Interpolation Method:**
    - Select **Linear**.
  4. **Define the Extent:**
    - Click on the **Extent slider** → **Calculate from Layer** → Select **Boundary2004\_550\_stpl83**.
  5. **Set the Output Raster Size:**
    - **Pixel Size X:** 5
    - **Pixel Size Y:** 5
  6. **Save the Interpolated Raster File**, then click **Run** and **Close**.
  - 7.

This step creates a **Triangulated Irregular Network (TIN) interpolation**, generating a **continuous elevation surface** from the provided elevation points.

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Interpolation - TIN Interpolation

ParametersLog

Input layer(s)

Vector layerIslands\_2004\_550\_stpl83

Interpolation attribute1.2 Elevation

☐ Use Z-coordinate for interpolation

Vector layerAttributeType

Arlington\_So...ELEVATIONPoints

Islands\_2004...ElevationBreak lines

Interpolation method

Linear

Extent

2357992.2101,2370654.1407,6928567.1592,6950323.1460 [EPSG:31433]

Output raster size

Rows4352Columns2533

Pixel size X5.000000Pixel size Y5.000000

Interpolated

C:/Users/jaymh/OneDrive/Desktop/elevation\_tin.tif

☒ Open output file after running algorithm

TIN interpolation

Generates a Triangulated Irregular Network (TIN) interpolation of a point vector layer.

With the TIN method you can create a surface formed by triangles of nearest neighbor points. To do this, circumcircles around selected sample points are created and their intersections are connected to a network of non overlapping and as compact as possible triangles. The resulting surfaces are not smooth.

The algorithm creates both the raster layer of the interpolated values and the vector line layer with the triangulation boundaries.

0%

Cancel

AdvancedRun as Batch Process...

RunCloseHelp

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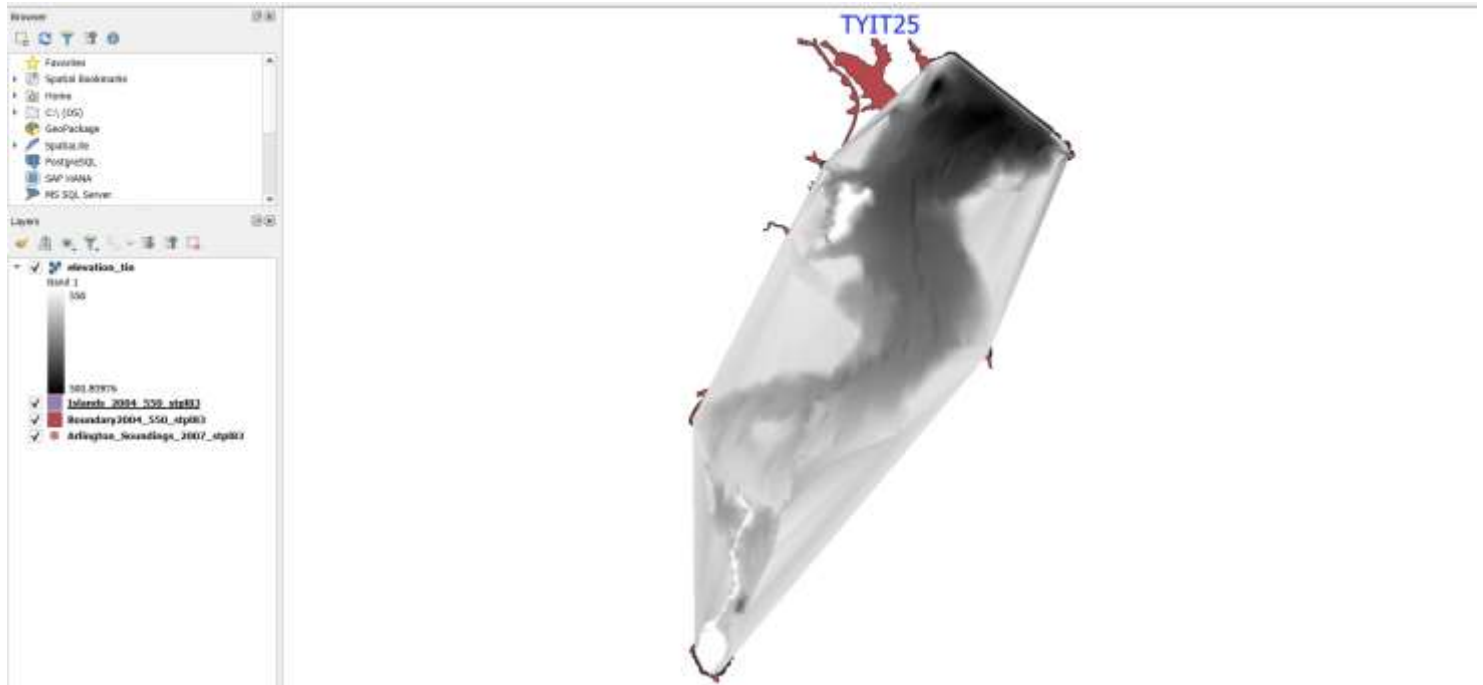
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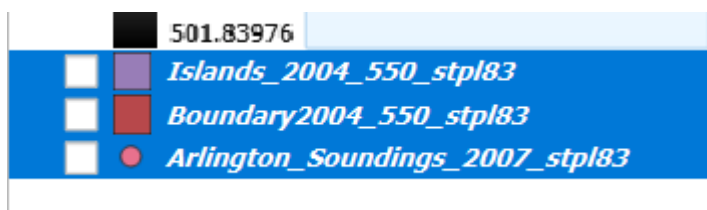
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### Step 17: Deselecting Layers

1. In the **Layers Panel**, locate the following layers:
  - **Islands\_2004\_550\_stpl83**
  - **Boundary2004\_550\_stpl83**
  - **Arlington\_Soundings\_2007\_stpl83**
2. **Uncheck** the checkboxes next to these layers to **deselect** or hide them from the map view.

This step helps focus on the interpolated raster layer by temporarily removing unnecessary vector layers from the display.



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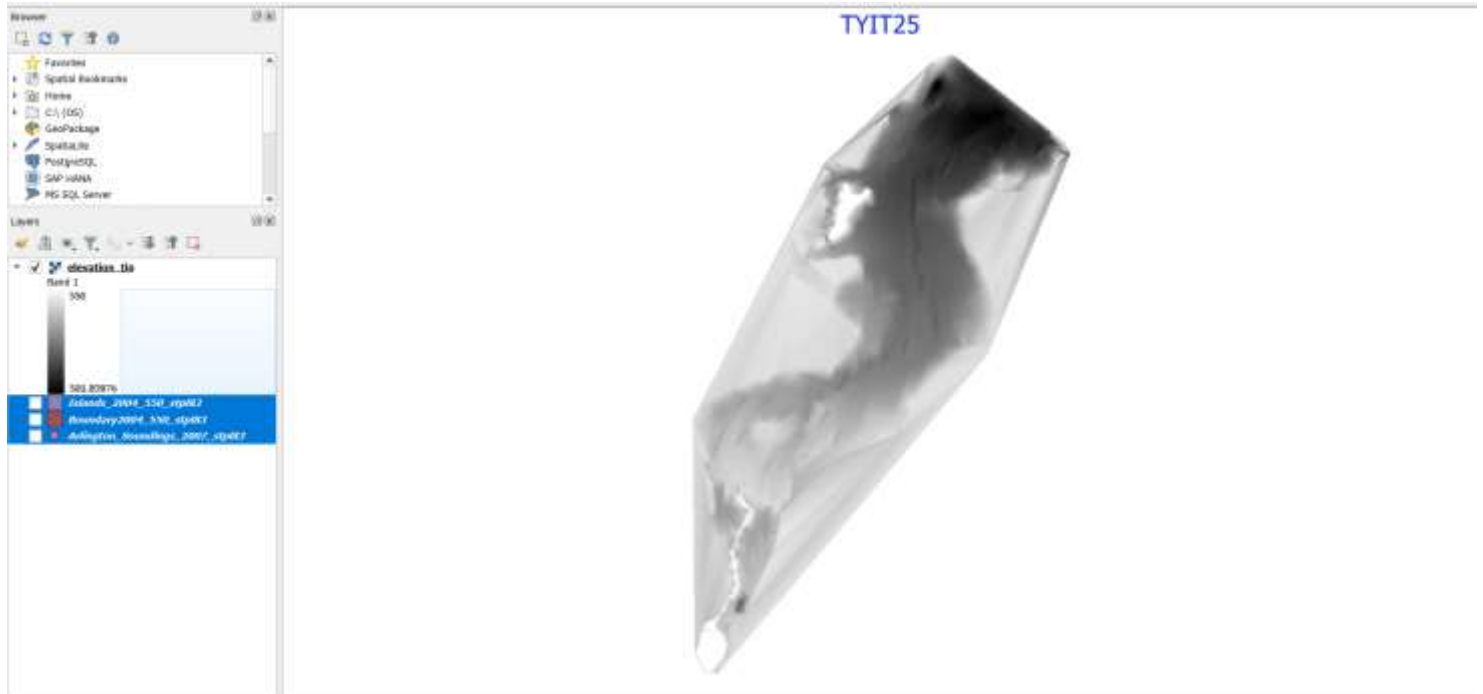
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### Step 18: Clipping Raster by Mask Layer

1. Navigate to **Processing** → **Toolbox** → **GDAL** → **Raster Extraction** → **Clip Raster by Mask Layer**.
2. **Set the input parameters:**
  - **Mask Layer:** Select **Boundary2004\_550\_stpl83 [EPSG:2276]**.
  - **Clipped Mask:** Save the output file to the desired location.
3. Click **Run** and then **Close**.

This step **clips the interpolated raster** to match the boundary defined by **Boundary2004\_550\_stpl83**, ensuring the raster is limited to the area of interest.

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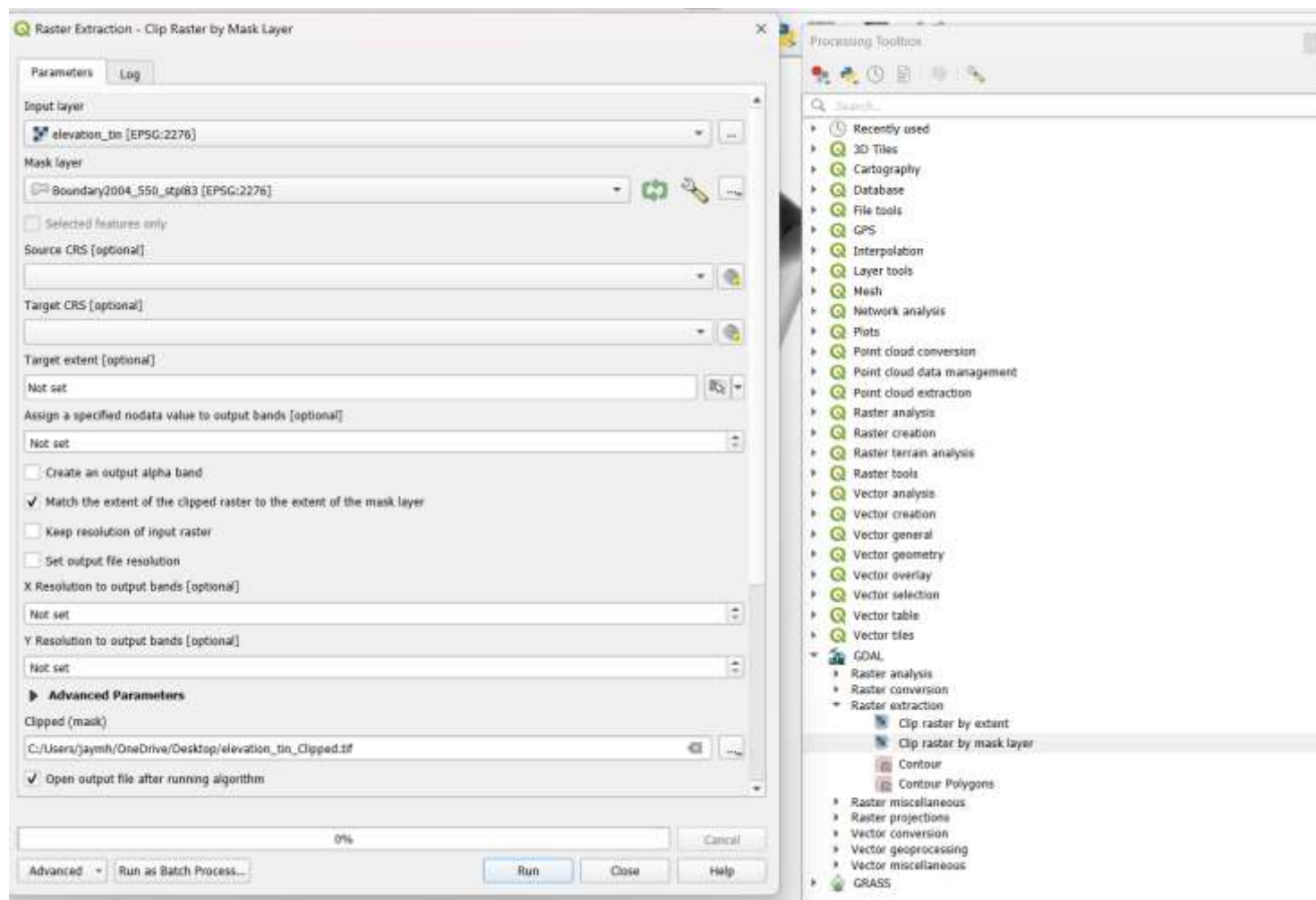
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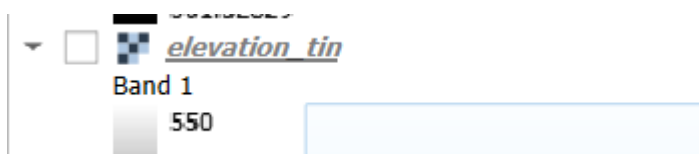
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### Deselecting the Previous Layer

1. In the **Layers Panel**, locate the layer named **elevation\_tin**.
2. **Uncheck** the checkbox next to it to **deselect** or hide the layer from the map view.

This step ensures that only the **clipped raster** remains visible for further analysis or visualization.



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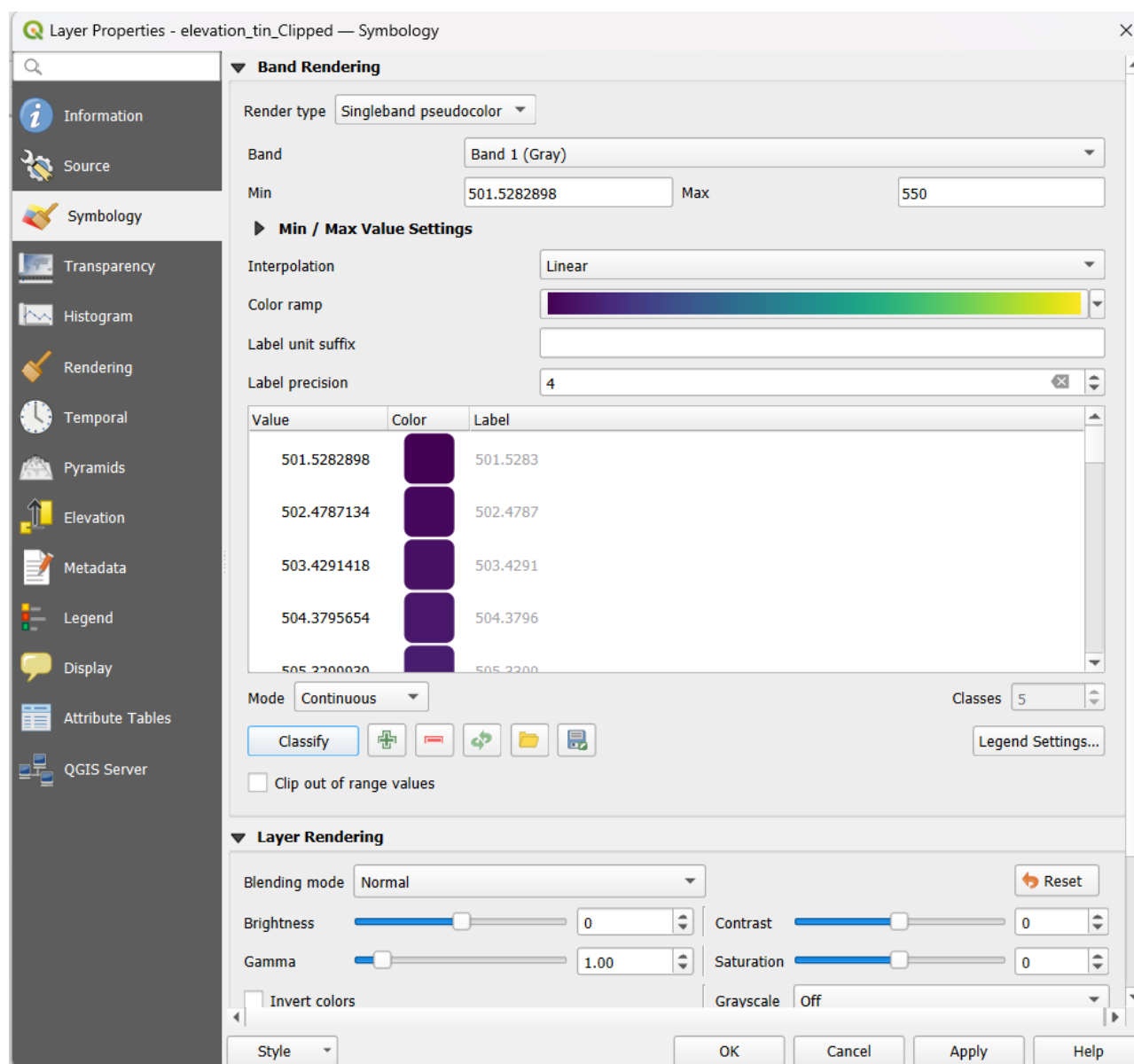
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### Step 19: Applying Symbolology to Clipped Elevation Layer

1. Double-click on **elevation\_tin\_Clipped** in the **Layers Panel** to open its **Properties** window.
2. Navigate to the **Symbolology** tab.
3. Set the rendering type:
  - In **Render Type**, select **Singleband pseudocolor**.
4. Customize the color map:
  - Choose a **color gradient** based on your preference.
5. Click on **Classify** to generate the color distribution.
6. Click **Apply** and then **OK** to finalize the changes.

This step enhances the **visual representation** of the elevation data, making it easier to interpret variations in terrain elevation.



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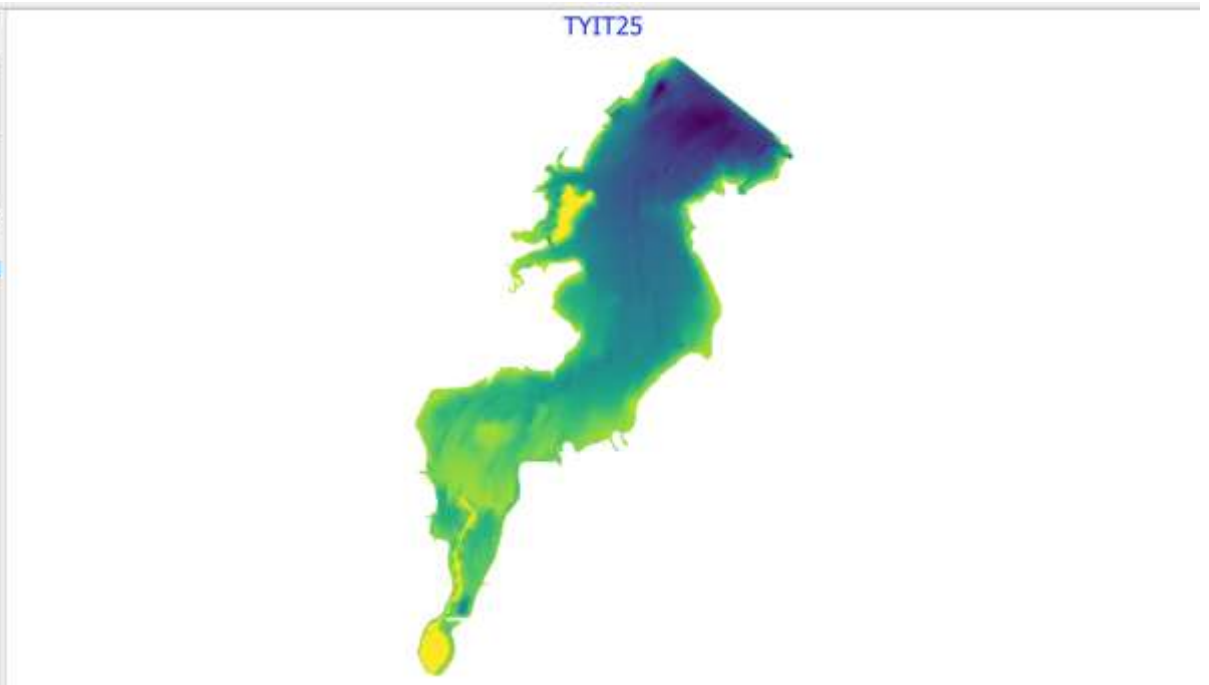
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### Step 20: Generating Contour Lines from Clipped Elevation Data

1. Navigate to **Processing** → **Toolbox** → **GDAL** → **Raster Extraction** → **Contour**.
2. **Set the input parameters:**
  - **Input Layer:** Select **elevation\_tin\_Clipped [EPSG:2276]**.
  - **Interval between contour lines:** Set to **5.000000**.
  - **Contours:** Save the output file to the desired location.
3. Click **Run** and then **Close**.

This step **creates contour lines** at 5-meter intervals, helping visualize elevation changes and terrain variations.

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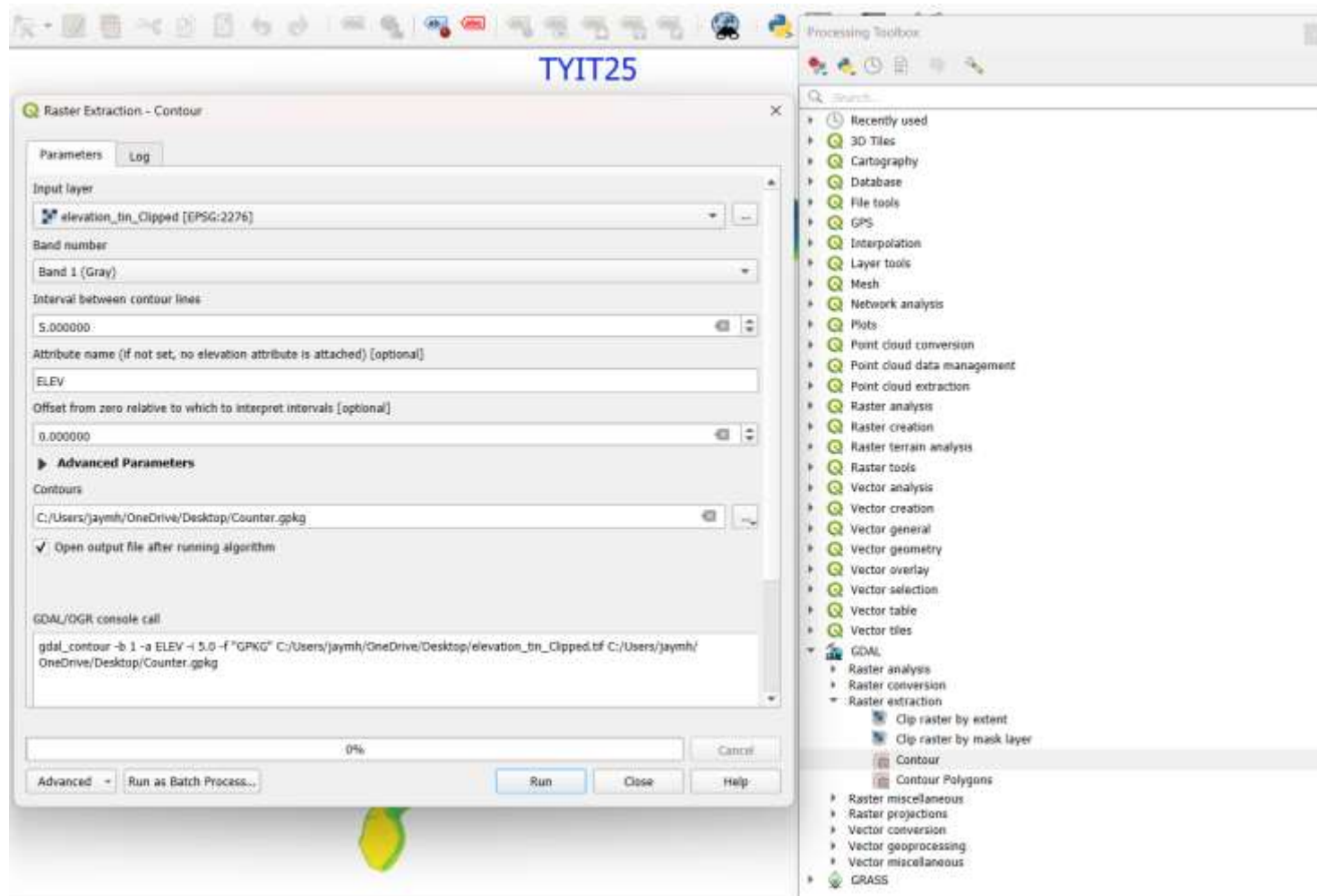
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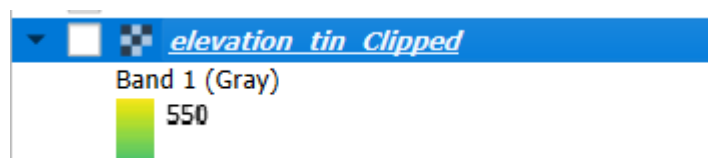
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### Deselecting the Clipped Elevation Layer

1. In the **Layers Panel**, locate the layer named **elevation\_tin\_Clippped**.
2. **Uncheck** the checkbox next to it to **hide** the layer from the map view.

This step ensures that only the **generated contour lines** remain visible for better visualization of elevation changes.



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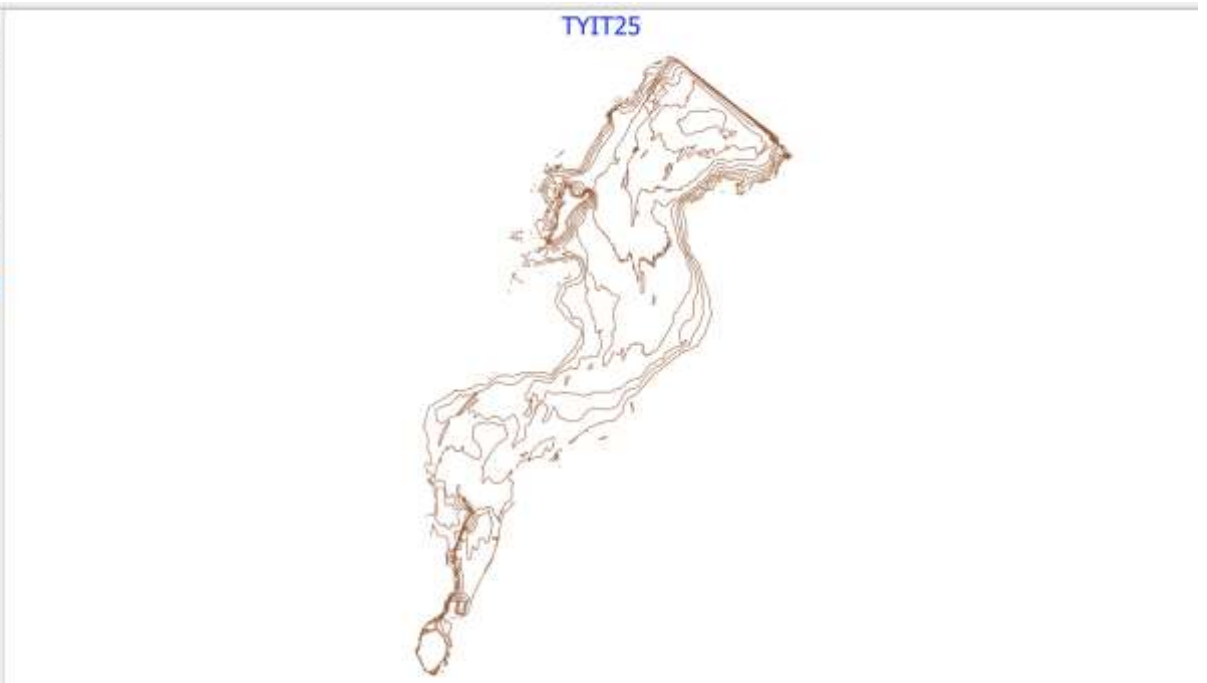
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### Step 21: Labeling Contour Lines

1. **Right-click** on the **Contour** layer in the **Layers Panel** and select **Properties**.
2. Navigate to the **Labels** tab.
3. **Enable labeling**:
  - Select **Single Labels**.
  - In **Value**, choose **ELEV** (Elevation attribute).
4. **Adjust placement settings**:
  - Click on **Mode** and select **Curved** to align labels with contour lines.
5. Click **Apply**, then **OK**.

This step ensures that elevation values are **clearly displayed** along the contour lines, enhancing map readability.

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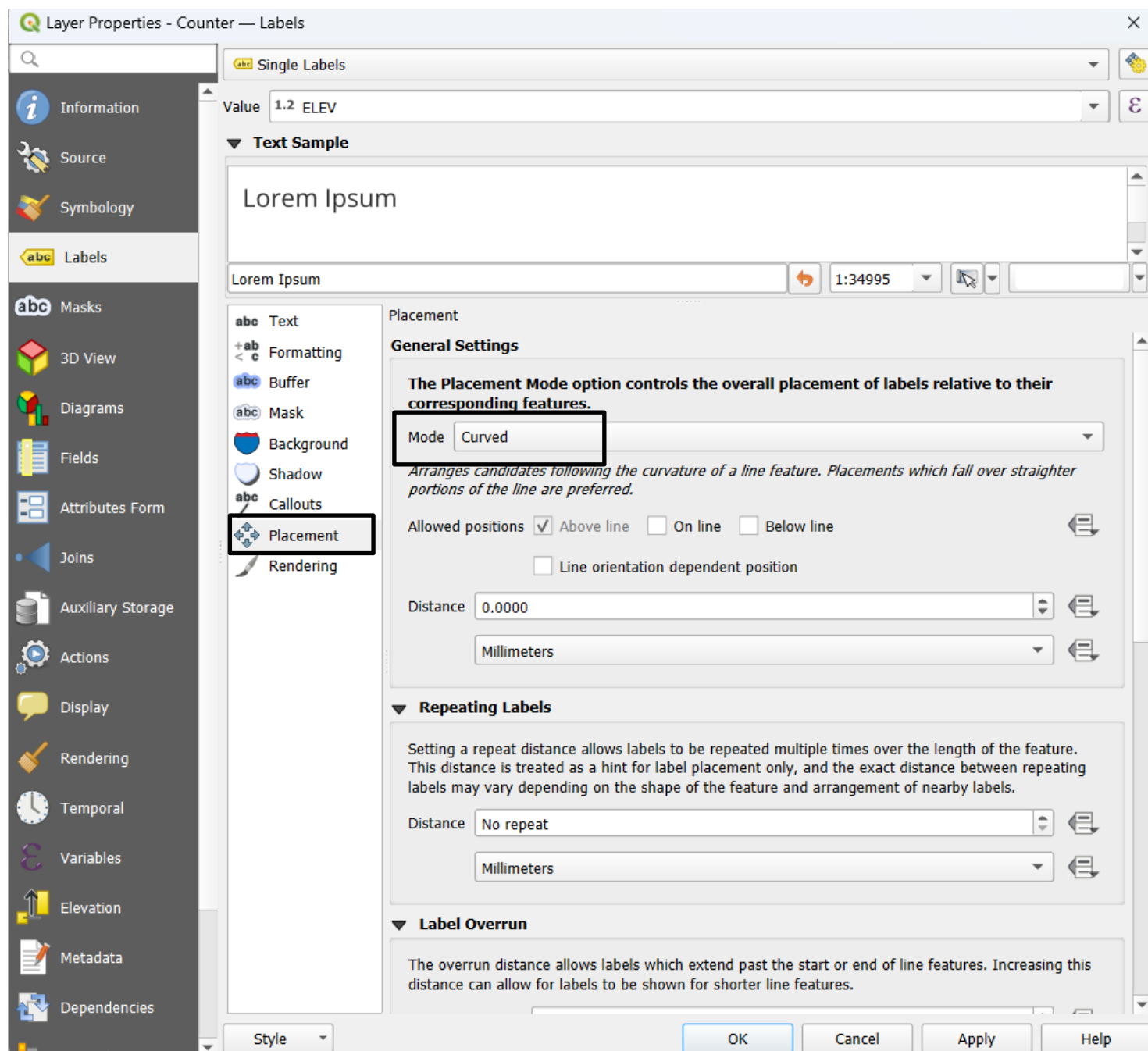
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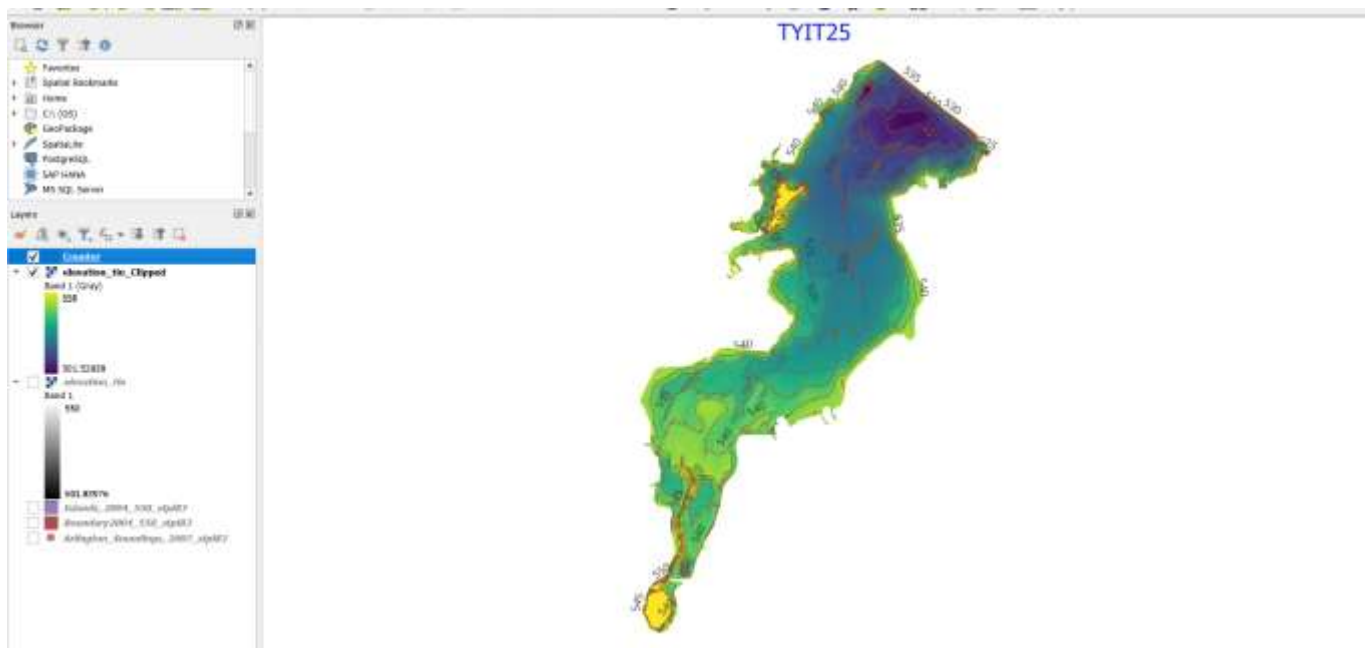
### Selecting the Clipped Elevation Layer

1. In the **Layers Panel**, locate the **elevation\_tin\_Clip** layer.
2. **Check the box** next to it to make it visible again.

This step ensures that both the **contour lines** and the **clipped elevation raster** are displayed together, providing a **clear and detailed** visualization of the terrain.

### FINAL OUTPUT 8C:

- Elevation raster (TIN Interpolation) is visible.
- Contour lines with elevation labels are displayed.
- A complete terrain analysis is achieved.



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